A Global 3° Periodic Structure in the UNESCO World Heritage Longitude Distribution: Domed Monuments, the Babylonian Beru, and a Levantine Harmonic Grid

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Code and data: github.com/SethTenenbaum/gerizim-paper-a

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Abstract

This exploratory study (no timestamped pre-registration) documents a global 3° periodic structure in the UNESCO World Heritage Cultural and Mixed longitude distribution ($N = 1011$).

Primary result — corpus-level periodicity. Every longitude of the form x.18°E, all 120 such anchors equally spaced around the globe, yields the maximum observed count of sites within 6.7 km of a harmonic node (Tier-A+, against a 4% geometric null). The spacing between consecutive x.18°E anchors is exactly 3° = 0.1 beru, the smallest integer sub-division of the Babylonian angular unit (MUL.APIN, c. 1000 BCE). A permutation check confirms that the phase concentration of the 56 Tier-A+ sites is not reproduced by random draws from the full-corpus longitude distribution ($p_{\text{perm}} < 0.0001$; the A+-only Rayleigh R is structurally near-perfect by construction — see §4.3 — and is reported as a description of A+ classification geometry, not as independent evidence for the periodicity).

Levantine corridor and circularity. Gerizim's longitude falls within the x.18°E phase band; this coincidence initiated the analysis and all results are treated as exploratory throughout (§1.1).

Sub-population and cluster results. Domed and spherical monuments ($N = 90$, raw morphological keyword filter) show 12.2% Tier-A+ against the 4% null (permutation $Z = 3.20$, $p = 0.004$; Bonferroni $p_{\text{adj}} = 0.004$, $k = 2$ pre-specified tests; conservative $k = 5$ bound: $p_{\text{adj}} = 0.005$ ***). A+-bearing harmonics host 4.19× more total sites than non-A+-bearing harmonics (permutation $p = 0.0033$;

$k = 5$ bound: $p_{\text{adj}} = 0.017$ **). Leave-one-out checks confirm no single site drives either result.

Alternative explanations. Geographic concentration of Eurasian civilizations in longitude bands that happen to align with beru harmonics cannot be excluded on this corpus alone. A sharpness permutation test shows the ±0.3% sensitivity collapse is *not* unusual under geographic concentration ($p_{\text{perm}} = 0.1728$, ns). A stronger specificity test asks whether the peak lands specifically at the metrologically attested 0.1000-beru spacing: 0.1000 beru ranks #1 of 7 by combined $-\log_{10}(p)$ in the observed data, and the fraction of permutations where it is the unique best spacing is $p = 0.1728$ (ns; §5.2). Unit sensitivity is therefore consistent with, but does not independently confirm, the metrological hypothesis. The primary evidence against a purely geographic account comes from the dome enrichment surviving all three simulation nulls and exceeding the expectation from monuments at the same longitude footprint (Null C, §4.1). A residual ambiguity remains: Null A confirms that domed monuments' own longitudes encode the enrichment, so geographic-concentration remains a viable first-level explanation (§1.1). The primary contribution is documentation of a reproducible quantitative regularity; the metrological interpretation awaits independent historical confirmation.

Keywords: archaeoastronomy, ancient metrology, geodetic alignment, UNESCO World Heritage, Babylonian beru, history of astronomy, statistical archaeology, spatial analysis, Mount Gerizim

1 Introduction

What this paper reports. The UNESCO World Heritage Cultural and Mixed corpus ($N = 1011$) appears to contain a global 3° periodic structure in its longitude distribution. Every longitude of the form $x.18$°E — all 120 such anchors equally spaced around the globe — produces the maximum observed Tier-A+ count when used as an angular reference (sites within $\lesssim 6.7$ km of a harmonic node, against a 4% geometric null). The 3° spacing equals the smallest integer sub-division of the Babylonian *beru* unit (0.1 beru = 3° = 3 *uš*, MUL.APIN, c. 1000 BCE). The analytic and interpretive consequences of the Gerizim phase-coincidence that initiated this study are defined in §1.1.

1.1 Circularity Containment Protocol

This subsection is the single canonical statement of the circularity problem and its analytic consequences. All later mentions in the paper are back-references to this section.

How the circularity arose. This study began with the hypothesis that the Babylonian beru unit might be detectable as a longitude-spacing signal in the UNESCO corpus, using Mount Gerizim (35.269°E) as the reference anchor. Gerizim was chosen on the basis of its documented *tabbur ha-aretz* ("navel of the earth") role in Samaritan cosmography (Purvis, 1968; MacDonald, 1964; Crown, 1989) and because the Levant is the natural geographic starting point for any study linking Babylonian metrology to the ancient world. Running the global anchor sweep subsequently revealed that Gerizim's longitude falls within the $x.18$°E phase band. The discovery that Gerizim's longitude aligns with the optimal phase, and the decision to frame this paper around a Levantine corridor, are therefore not fully separable events. If the sweep had revealed that the optimal phase was, say, $x.55$°E — far from Gerizim's 2.269° mod 3.0° — this paper would not exist in its current form. The fact that Gerizim's longitude falls within the optimal phase band is what initiated the analysis and shaped every framing decision.

What the circularity contaminates (full list).

- The Levantine corridor statistics (Tests 1, 2, 3 as computed from Gerizim): the decision to report them was conditioned on observing the phase alignment.
- The $x.18$°E framing and label: this phase band was identified in the sweep, not predicted independently.
- The Rayleigh Test B: the near-perfect R is a mathematical consequence of the A+ classification criterion (sites classified as A+ by definition land within ±0.006° of the $x.18$° phase); it does not independently confirm the periodicity.
- All narrative emphasis on the Levantine corridor as historically motivated: the motivation is genuine, but the motivation and the result are not independent observations.

What the circularity does *not* contaminate (full list).

- *The existence of a 3° periodic structure in the UNESCO longitude file.* This is a reproducible property of the coordinate data: any investigator running an unanchored 120-phase scan will recover it without reference to Gerizim.

- *The morphological specificity of domed monuments (Test 2).* The dome enrichment does not depend on the identity of the anchor, only on the beru spacing and tier threshold, both of which were fixed on metrological grounds before any beru distances were computed (§3.2). All 120 x.18°E anchors produce identical A+ counts for the dome population; Gerizim is not special.

- *The cluster asymmetry (Test 3).* This is a full-corpus result requiring no keyword filter and no anchor beyond the 0.1-beru grid itself.

- *The unit sensitivity sweep.* The sharp collapse at ±0.3% of the canonical spacing is a property of the corpus structure, not of the choice of Gerizim as anchor. (Its interpretation is still limited by the permutation test result; see §5.2.)

The protocol. *This identification problem cannot be closed by any statistical method applied to this corpus*, because every result was computed and reported after the phase alignment was observed. The protocol applied throughout this paper is therefore:

(a) All results are treated as *exploratory* without exception. No result is described as confirmatory.

(b) The Bonferroni correction is applied as a *descriptive bound* on effect magnitude, not as a frequentist guarantee of false-positive control.

(c) The Levantine corridor statistics are reported as the motivated entry point into the x.18°E structure, not as independent evidence for it.

(d) Every downstream section that touches the circularity back-references this section rather than re-stating the argument.

Scope of the x.18°E label. "x.18°E" denotes the optimal phase band of the 3° periodic structure as identified in the anchor sweep. It is a rounded shorthand, not an independently derived prediction. Any 120-anchor phase-scan of the UNESCO longitude distribution would recover this structure; the label summarises it.

Why this design avoids the main pitfalls of archaeoastronomy. Prior investigations of metrological patterns in monumental architecture have been rightly criticized because investigators assembled the very site lists they then tested (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). Three design features distinguish this study from that tradition. First, the site list (the UNESCO World Heritage corpus) was compiled by UNESCO without reference to this hypothesis and is fixed; it cannot be modified by the author. Second, the angular interval tested (0.1 beru) was selected on metrological grounds (§3.2), not tuned post-hoc to maximize the result. Third, the tier boundaries and the sub-population keyword sets were specified before any beru distances were computed. A genuine circularity does remain; its scope and consequences are defined exhaustively above (§1.1).

Structure of the paper. Section 2 provides background on the beru unit and the geodetic framework. Section 3 describes data, tier classification, test populations, and the multiple-comparisons protocol. Section 4 presents results in test order. Section 5 reports robustness checks. Section 6 catalogues limitations. Section 7 evaluates alternative explanations and outlines future work. The Appendix contains the keyword audit; detailed unit-sweep tables, the corridor ranking, and the founding-classifier analysis are deposited as supplementary material.

2 Background

2.1 The Beru Unit

The *beru* (30° of arc) is attested in the MUL.APIN cuneiform compendium (Hunger & Steele, 2019; Watson & Horowitz, 2011), the composition of which is dated on broad grounds to the late second or early first millennium BCE (a conservative range; the canonical tablet copies date to c. 687 BCE, but internal astronomical content points to original observation dates of c. 1000–1400 BCE; see Hunger & Steele 2019, pp. xi–xv). In this system, the unit divided the ecliptic into twelve equal segments and served for both stellar and terrestrial angular measurement. The terrestrial use is attested explicitly in MUL.APIN II ii 43–iii 15 (the "shadow-length" table), which converts beru of daylight into ground distances (Hunger & Steele, 2019, pp. 2–15). The sub-division of the beru into 30 *uš* (1 uš = 1°) is well attested (Hunger & Steele, 2019; Robson, 2008), and the 0.1-beru interval (3°) tested here corresponds to one-tenth of a *beru*, or exactly 3 *uš*, an integer number of the smallest standard angular unit.

The beru also functioned as a unit of territorial description in administrative and cosmological contexts beyond astronomical tables. Royal inscriptions and administrative texts record inter-regional distances in beru between major cities and outlying territories (*nagû*) (Neugebauer, 1975; Robson, 2008). On the Babylonian *Imago Mundi* (the so-called Map of the World), the distances between the "islands" beyond the encircling Salt Sea are explicitly marked as multiples of beru, placing the unit in a cosmographic as well as a practical geographic register. The parallel institution of the *kudurru* (boundary stone) records land-grant limits and territorial extents in cuneiform inscriptions, and many kudurru were deposited in temples as legally authoritative records (Robson, 2008; Powell, 1987). The combination of a fractional angular unit employed at cosmological scales and a tradition of temple-deposited territorial records is the documentary basis for testing whether a 0.1-beru harmonic spacing is detectable in the longitude distribution of major monuments. This study tests the statistical pattern; the mechanism by which any such correspondence could arise is not documented and is outside the scope of this paper.

2.2 Ancient Metrology and Geodetic Practice

Whether pre-modern societies employed standardized angular measures across large geographic scales has a long and contested scholarly history. Thom's "megalithic yard" (Thom, 1967, 1971, 1964) stimulated statistical refinement (Kendall, 1974) and methodological critique. Freeman showed Thom's quantum analysis was sensitive to selection effects (Freeman, 1976), and Kendall cautioned that apparent quanta can arise from rounding in the source data (Kendall, 1974). Powell's treatment of Mesopotamian metrology (Powell, 1987) remains the standard reference for the system within which the beru is embedded. More recently, Magli (Magli, 2013, 2016) has applied Monte Carlo permutation methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles, 1999, 2015; Silva, 2020). The ICOMOS–IAU thematic study (Ruggles & Cotte, 2010) provides the institutional context for archaeoastronomical analysis of World Heritage properties, and Aveni (Aveni, 2001, 2003) and Neugebauer (Neugebauer, 1975) supply the broader disciplinary framework.

The present study builds on this tradition but departs from it in three respects. First, the site list is externally curated (the UNESCO World Heritage List), eliminating the selection bias that has undermined many archaeoastronomical claims. Second, the test is

one-dimensional (longitude only), yielding a well-defined geometric null expectation (4%) with no free parameters. Third, the analysis is fully computational and publicly archived, permitting exact reproduction (Tenenbaum, 2026).

Prior investigations of ley-line hypotheses (Watkins, 1925; Williamson & Bellamy, 1983; Hutton, 1991) were rightly criticized for post-hoc site selection and unconstrained geometric degrees of freedom (Kendall, 1974). The present design avoids all three problems: the site list is frozen before analysis, the test statistic is established in advance on metrological grounds, and the geometric null is analytically derived.

2.3 The Anchor: Mount Gerizim

The circularity arising from Gerizim's longitude coinciding with the optimal phase band is defined exhaustively in §1.1 and is not restated here. The present subsection documents only the geodetic and textual rationale for the anchor choice.

Mount Gerizim (35.269°E) is used as the primary reference anchor because of its documented *tabbur ha-aretz* ("navel of the earth") role in Samaritan cosmographic tradition (Purvis, 1968; MacDonald, 1964; Crown, 1989), a role analogous to that of Babylon's meridian in cuneiform astronomical records (Hunger & Steele, 2019). This tradition predates the major world religions whose sites appear in the UNESCO corpus and provided the textual basis for the initial hypothesis. No optimization across alternative Levantine anchors was performed.

Gerizim's tentative-list coordinate (ref. 5706, DMS E35 16 08) is externally sourced and does not appear in the UNESCO XML inscribed corpus, eliminating one form of circularity by construction (the anchor cannot also be a test point). The coordinate is taken from Palestine's UNESCO tentative list (2012); because tentative-list pages can be updated, an archived copy of the source page has been deposited in the public repository (https://github.com/SethTenenbaum/gerizim-paper-a). Jerusalem (35.2317°E), approximately 4.1 km east of Gerizim, is an inscribed corpus member; it is self-excluded when used as anchor and produces the same A+ count from the same corridor (§5.3). The enrichment therefore belongs to the 4.1 km Gerizim–Jerusalem corridor, not to the specific choice of anchor within it. Mecca reaches only $p \approx 0.057$ (ns) and is not in the signal corridor.

2.4 Interpretive Framework

This paper is a pattern-detection study, not a historical attribution. Three classes of explanation are evaluated throughout.

1. **Geographic concentration.** Major civilizations that produced UNESCO-ranked monuments developed in Eurasian longitude bands that align with 0.1-beru harmonics from the Levant. The simulation null models (§5.5) preserve the empirical longitude distribution; the dome sub-population survives all three, confirming the signal is not simply a consequence of site density at any given longitude. However, the dedicated geographic-concentration simulation (§4.1) shows that the dome sites' own longitude positions reproduce the observed count (Null A, ns), so geographic concentration remains a viable first-level account.

2. **Metrological transmission.** A 3° (0.1-beru) angular reference system anchored near the Levant informed the placement of major foundations through trade, imperial administration, or religious networks. The morphological concentration in domed

forms and the cluster asymmetry are consistent with this reading. No mechanism is documented; the interpretation is a motivated hypothesis.

3. **Unmodeled statistical artefact.** A systematic bias in UNESCO curation, geocoding, or the discrete longitude distribution produces the apparent periodicity. An artefact account must additionally explain the morphological specificity of the dome enrichment, the harmonic cluster asymmetry, and why the signal is strongest in the oldest inscription cohorts — none of which are predicted by any known curation or geocoding bias.

The study documents the pattern and the evidence bearing on each explanation. Adjudicating between explanations 1 and 2 requires independent corpora beyond the scope of this paper.

3 Data and Methods

3.1 Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E, from Palestine's UNESCO tentative list (ref. 5706, DMS E35 16 08, 2012). Gerizim is not on the inscribed World Heritage List and does not appear in the test-site corpus, avoiding circularity by construction. In each sweep the anchor is excluded from the test population; Jerusalem is also excluded in both sweeps (§2.3).

For a site at longitude λ, the beru-distance is $B = |\lambda - 35.269|/30$ and the beru-deviation from the nearest 0.1-beru harmonic is $\delta = |B - \mathrm{round}(B, 0.1)|$. Longitude differences are in angular measure; the tier classification is therefore a fixed angular criterion, not a fixed physical radius. The kilometre labels reported alongside tier thresholds (e.g. $\lesssim 6.7$ km for Tier-A+) are approximate equatorial equivalents computed as $\delta \times 30 \times 111$ km/° and become smaller at higher latitudes: a site at 60°N that satisfies $\delta \leq 0.002$ beru is physically within $\sim 55.7/111 \times 6.7 \approx 3.4$ km east-west of the harmonic. The Gerizim–Jerusalem separation of 4.1 km is likewise an equatorial approximation (no cosine-latitude correction); at 32°N the true east-west distance is approximately $4.1 \times \cos(32°) \approx 3.5$ km. This convention is applied uniformly throughout the study so that all reported kilometre values are directly comparable. This makes the angular criterion *conservative* relative to a fixed physical-km threshold at high latitudes. A geodesic sensitivity check (east-west haversine distances) confirms this: 0 of the 56 current A+ sites would be removed under a strict 6.7 km physical threshold, while 14 additional high-latitude sites would qualify, increasing the A+ count to 70 and strengthening all primary results. No correction for latitude is therefore applied; the reported km figures are equatorial approximations provided for descriptive context only. The geometric null for Tier-A+ is $2 \times 0.002/0.1 = 4\%$. This figure assumes uniformly distributed longitudes. It is used throughout as a convenient reference baseline, but the primary tests against the chance explanation are the simulation-based null models (§5.5), which preserve the full empirical longitude distribution and do not rely on this assumption. A Gaussian KDE fitted to the observed longitude distribution produces 40.4 ± 6.2 expected A+ out of 1011 sites ($\sim$4.0%), confirming the geometric null is not materially inflated by longitude clumping, and that binomial results calibrated to 4% are reasonable approximations (§5.5).

Throughout this paper, exact binomial p-values are reported as a transparent reference baseline against the geometric null. The *primary* tests against geographic structure are

the simulation-based null models (§5.5), including longitude permutation, bootstrap, and KDE-smoothed null, which preserve the empirical longitude distribution and do not assume i.i.d. sampling. Where simulation and binomial p-values diverge, the simulation figure takes precedence. Binomial p-values are best understood as convenient approximations that assume uniform longitude spacing. Because this assumption is violated by the UNESCO corpus, the simulation models are designed to relax it.

3.2 Metrological Basis for the 0.1-Beru Spacing

The harmonic spacing tested throughout this paper is 0.1 beru (3°). This interval is the smallest integer sub-division of the beru attested in cuneiform metrology: it equals exactly 3 *uš* (3°), an integer multiple of the smallest standard angular unit in the sexagesimal system (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). It is therefore not an arbitrary spacing but a metrologically motivated candidate.

The only ancient angular units of which the author is aware that are metrologically attested in cuneiform records and applicable to geographic arc are the Babylonian beru (30°) and its sexagesimal sub-units. The Vedic Hora and Rashi are structurally identical at the sub-divisions tested here (both yield the same grid); they are not alternatives to the beru but the same angular framework in a different textual tradition. The 0.1-beru interval (3°) was selected because it is the finest metrologically attested integer sub-division that yields a tractable test (120 harmonics in 360°, 4% geometric null). Coarser spacings (0.5 beru, 1 beru) have geometric null rates of 50% and 100% and are uninformative; finer sub-divisions (0.02 beru) were tested and do not produce the signal (Table 10, §5.1). No spacing was selected post-hoc to maximize the result.

The 0.05-beru spacing (6°) also reaches significance in both populations (Table 10). This is expected on structural grounds and does not constitute an independent signal: every 0.10-beru harmonic is by definition also a 0.05-beru harmonic (since $0.10 = 2 \times 0.05$), so a site that falls within 6.7 km of a 0.10-beru node automatically falls within 6.7 km of a 0.05-beru node as well. The 0.05-beru result is therefore a coarser sub-sampling of the same underlying 0.10-beru periodicity, not a separate signal. The 0.10-beru spacing produces stronger joint significance than 0.05-beru (Table 10) and is preferred on both metrological (smallest attested integer sub-unit) and statistical (sharpest peak) grounds; the 0.05-beru result is a predictable mathematical consequence of it.

The anchor and unit sensitivity sweeps (§5.3, §5.1) probe the same underlying x.18°E periodic structure from two angles and should not be counted as independent lines of evidence. Independent evidence comes from morphological specificity (the dome enrichment surviving simulation nulls) and the cluster asymmetry. Both the anchor sweep and the unit sensitivity observation are consistent with the metrological hypothesis. A sharpness permutation test shows the ±0.3% collapse is not unusual under geographic concentration alone ($p = 0.1728$, ns; §5.2). A stronger specificity test (§5.2) asks whether the peak lands specifically at 0.1000 beru: $p = 0.1728$ (ns). Neither robustness check independently discriminates the two explanations beyond the primary dome-enrichment result.

3.3 Tier Classification (fixed before data analysis)

Sites are classified by δ, the angular deviation from the nearest 0.1-beru harmonic (interval $\approx 3°$). All tier boundaries and the six test populations were defined and frozen before any beru values were computed. The corpus was not compiled by the author and cannot be

modified.

- **Tier A++**: $\delta \leq 0.0002$ beru ($\lesssim 0.67$ km at equator). $P_0 = 0.4\%$. *Descriptive only; not tested or reported as evidence.* A++ counts appear in the anchor sweep summary table (Table 12) for descriptive reference to sub-kilometre precision alignments. No formal test is performed on this tier.
- **Tier A+**: $\delta \leq 0.002$ beru ($\lesssim 6.7$ km at equator). $P_0 = 4\%$. Primary test threshold.[1]
- **Tier A**: $\delta \leq 0.010$ beru ($\lesssim 33$ km at equator). $P_0 = 20\%$.
- **Tier B**: $\delta \leq 0.050$ beru.

3.4 Domed and Spherical Monuments (Test 2)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV web-page text. All 7 keywords are treated identically (positive whole-word match, no context gating), making the population mechanically determined by the keyword list. Final population: $N = 90$. Primary tests: simulation-based null models (permutation, bootstrap, KDE) reported in §5.5; exact binomial A+ ($> 4\%$) and A ($> 20\%$) reported as transparent reference baselines; χ^2 uniformity and anchor sweep also reported.

A context-validated robustness check (Test 2x) applies sentence-level architectural confirmation. Gate 1 (28 negative patterns) rejects geological, vernacular, and non-architectural uses, while Gate 2 (86 positive patterns) requires architectural confirmation. This reduces the population to $N = 83$ (7 sites rejected, e.g. Uluru "rock domes," Hiroshima "Genbaku Dome," full list in Appendix A.1). The validated version yields $p = 0.0005$, $3.31\times$ enrichment, which is stronger than the raw sweep ($p = 0.0009$, $3.06\times$), consistent with the 7 rejected sites being non-architectural false positives that dilute rather than drive the signal. Because the validation rules were developed after the initial keyword sweep, Test 2x is a post-hoc robustness check and is excluded from the primary Bonferroni family. The raw-sweep result is the conservative primary figure.

3.5 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extends the core domed/spherical test (Test 2) by adding earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun, mound*). The categories are grouped on the basis of shared hemispherical or quasi-hemispherical geometry, a form widely associated with cosmological and funerary symbolism across cultures. They are not grouped on the basis of a claimed architectural lineage. The label "evolution" refers to a morphological sequence in the statistical test design (mound $\rightarrow$ stupa $\rightarrow$ dome as progressively formalized built expressions of the same form class); it is not a claim that any category is architecturally derived from another.

The primary version (Test 2b) applies a raw positive keyword sweep: all 132 Cultural/Mixed sites matching any keyword in {*tumulus, tumuli, barrow, barrows, kofun, mound, stupa, stupas, dagoba, chorten, tholos, dome, domed, domes, spherical*} are

[1] $2 \times 0.002/0.1 = 4\%$: window width divided by harmonic spacing. Km labels are equatorial approximations; see §3 for the latitude-sensitivity discussion.

included without context gating, for the same pre-specification reasons as Test 2. A context-validated version (Test 2bx, Exploratory) applies the same architectural gates as Test 2x, reducing the population to $N = 126$. The validated version yields $p = 0.0005$, 2.78× enrichment, which is stronger than the raw sweep ($p = 0.0009$, 2.65×), consistent with the extra sites in the raw sweep being diluting false positives rather than driving the signal.

Test 2b is classified as a *sensitivity variant* of Test 2 rather than an independent hypothesis. Its population shares the entire dome/stupa core ($N = 90$) with Test 2 and adds only 42 mound-stage sites, which show no significant enrichment on their own (6.2%, $p = 0.3012$, ns). Counting Test 2b separately in the Bonferroni family would add a test whose population is almost entirely overlapping with Test 2. Readers who prefer a stricter $k = 5$ family should note that the dome result ($p_{\text{adj}} = 0.004$ at $k = 4$, $p_{\text{adj}} = 0.005$ at $k = 5$) survives either way. Tests: exact binomial A+ on combined and per-stage populations.

3.6 Cluster Asymmetry (Test 3)

Full corpus ($N = 1011$), no keyword filter. If the beru-harmonic hypothesis is correct, ancient surveyors would have preferentially founded or selected significant sites at harmonic longitudes, and subsequent construction, urban growth, and heritage accumulation would have clustered around those initial seed sites. This predicts that Tier-A+ harmonics should host more UNESCO sites in total, not just more A+ sites, than harmonics without any A+ hit. For each site, count other UNESCO sites within ±0.3° longitude. Compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U, permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

3.7 Founding-Site Enrichment (Test E [Exploratory])

Two independent keyword analyses on the full corpus ($N = 1011$), no keyword filtering for the primary tests:

1. **World religion founding sites.** Keyword match for Buddhism, Christianity, Islam, Judaism, Hinduism, Zoroastrianism in site name/XML description.

2. **UNESCO founding language.** Sites where UNESCO uses "first capital," "birthplace of," "founded in," etc. in the XML description.

A context-aware keyword classifier (`lib/founding_filter.py`) was applied to all sites. This classifier was iteratively tuned after examining misclassified sites (full disclosure in §3.10). Because refinement occurred after data inspection, the enrichment result from this classifier should be treated as *Exploratory*. The inscription-year gradient and world-religion sub-tests do not depend on this classifier.

3.8 Temporal Gradient (Test 4)

Cochran-Armitage trend test on the full corpus ($N = 1011$), no keyword filter. A+ rate is tested for a monotonic decline as UNESCO inscription year increases (earliest inscriptions reflect greatest site importance). Three-cohort grouping (1978–1984, 1985–1999, 2000–2025) is primary; five-cohort grouping (1978–1984, 1985–1991, 1992–1999,

2000–2009, 2010–2025) is secondary. The three-cohort bins are formed by merging the two middle five-cohort bins (1985–1991 and 1992–1999) into a single 1985–1999 bin. Code: `analysis/unesco/temporal_gradient_test.py`.

Note on the regional composition confound. The founding canon (1978–1984) is concentrated in Europe and the Mediterranean corridor, the Eurasian heartland where the beru-harmonic signal is strongest. Later cohorts add proportionally more sites from Africa, the Americas, and Oceania, regions where the null hypothesis predicts lower A+ rates regardless of any metrological effect. A within-region diagnostic (`regional_temporal_gradient.py`) finds that 4 of 5 regions show the early greater than late direction, but 0 reach individual significance, and the diagnostic is underpowered at $N = 5$ regions. The Mantel-Haenszel region-adjusted OR is 1.582 ($p = 0.1101$). The temporal gradient is therefore treated as descriptive only; it indicates that UNESCO's earliest inscription cohort is concentrated in the very regions where the beru-harmonic signal is strongest, but does not establish that time itself, independent of geography, drives the pattern.

3.9 Multiple-Comparisons Protocol

All results are exploratory and the Bonferroni correction is a descriptive bound on effect magnitude only (§1.1).

Primary pre-specified family: Tests 2 and 3 ($k = 2$). The two tests that address structurally independent questions and whose populations and statistics were specified before any beru distances were computed are:

- **Test 2** (domed/spherical monuments; morphological keyword filter, $N = 90$): tests whether this morphologically defined sub-population is enriched above the 4% null.
- **Test 3** (cluster asymmetry; full corpus, no keyword filter): tests whether A+-bearing harmonics attract more total sites than non-A+-bearing harmonics.

These two tests address structurally different questions on different populations and are not dependent on each other. Test 2 asks whether a morphologically-defined sub-population (domed/spherical monuments, $N = 90$) is over-represented *within 6.7 km* of a harmonic node — a proximity question. Test 3 asks whether harmonics that happen to have at least one A+ site also attract more *total* UNESCO sites within a $\pm 0.3°$ longitude window — a longitude-clustering question about the full corpus. A result could be present in one and absent in the other: for example, the dome proximity enrichment could survive while harmonics showed no overall density asymmetry, or vice versa. That both pass the Bonferroni bound strengthens the overall pattern.

Note on partial dependence. Both tests are conditioned on the same 56 A+ site identities: Test 2 draws its A+ sites from the dome sub-population (a subset of the 56), while Test 3 uses the A+ site longitudes to identify A+-bearing harmonics in the full corpus. They are therefore not fully statistically independent despite addressing different questions. Readers who prefer a conservative $k = 5$ family (adding Tests 1, 2b, and 4 to Tests 2 and 3) should note that both primary results survive that stricter bound: dome $p_{\text{adj}} = 0.005$ (***) and cluster $p_{\text{adj}} = 0.017$ (**) at $k = 5$. The primary Bonferroni family remains $k = 2$ because Tests 1 and 3 share A+ sites (full dependence at the data level), Test 2b is a population-overlapping sensitivity variant of Test 2, and Test 4 is excluded on substantive grounds (§4.5). The $k = 2$ and $k = 5$ adjusted values are reported in parallel throughout so readers can apply whichever bound they consider appropriate.

The Bonferroni per-test reference threshold at $k = 2$ is $\alpha/2 = 0.025$. Both pass this bound: dome Bonferroni $p_{\text{adj}} = 0.004$ (***), cluster Bonferroni $p_{\text{adj}} = 0.013$ (**). Both also pass the conservative $k = 5$ bound: dome $p_{\text{adj}} = 0.005$ (***), cluster $p_{\text{adj}} = 0.017$ (**).

Test 1 (full corpus enrichment): descriptive precursor. Test 1 asks whether the full corpus ($N = 1011$) shows above-chance Tier-A+ enrichment. It is reported as a descriptive result, not a primary test, for two reasons. First, Tests 1 and 3 share the same 56 A+ sites: they are partially dependent, and including both in a Bonferroni family without accounting for that dependence would overstate the number of independent tests. Second, the full-corpus result (raw $p = 0.0103$, 1.38× enrichment) is the weakest of the three non-temporal findings and is best understood as the backdrop against which Tests 2 and 3 demonstrate morphological and spatial specificity. Test 1 is reported with its raw p and Bonferroni reference at $k = 2$ ($p_{\text{adj}} = 0.041$) for completeness; readers should note that this adjusted value does not enter the primary family calculation.

Test 4 (temporal gradient): descriptive only, outside the family. The Cochran-Armitage trend test (raw $p = 0.0481$) is reported as a descriptive observation, not a primary test, and is excluded from the multiple-comparisons family. The reason is substantive, not merely statistical: a within-region diagnostic shows the pooled gradient is largely explained by regional composition rather than time itself (4/5 regions concordant, MH OR = 1.582, $p = 0.1101$; §4.5). Retaining a test in the family whose signal is attributable to a known confound would give a misleading impression of the strength of the family. Test 4 is therefore reported in full in §4.5 but carries no independent evidential weight.

Test 2b (hemispherical evolution): sensitivity variant, not counted. Test 2b extends Test 2 by adding mound-stage keywords ($N = 132$). Its population shares the entire dome/stupa core with Test 2 and adds 42 mound-stage sites, which show no significant enrichment on their own (6.2%, $p = 0.3012$, ns). Counting it as a separate family member would add a nearly-overlapping test without independent evidential content. Test 2b is reported as a sensitivity check on Test 2.

Test E (founding-site classifier): fully post-hoc. The founding-site classifier (`lib/founding_filter.py`) was iteratively refined after examining misclassified sites (§3.10). The enrichment result ($p = 0.009$) is fully post-hoc and is reported only in Appendix A.2.

Summary table.

Table 1: Multiple-comparisons summary. The primary pre-specified family is Tests 2 and 3 ($k = 2$), the only tests that are both structurally independent and whose populations were specified before any beru distances were computed. Test 1 is a descriptive precursor (shares A+ sites with Test 3). Test 4 is excluded from the family because its signal is largely explained by regional composition (§4.5). Test B is excluded: high R is a mathematical consequence of the A+ classification criterion and does not constitute independent evidence (§4.3). Conservative $k = 5$ adjusted values are shown alongside $k = 2$ for both primary tests. All results are exploratory (§1.1). Bonferroni per-test reference threshold at $k = 2$: $\alpha = 0.025$.

Test	Population	Raw p	p_{adj} ($k = 2$)	p_{adj} ($k = 5$)	
Test 2 (dome/spherical A+)	$N = 90$	0.0009	0.004 ***	0.005 ***	**Prima**
Test 3 (cluster asymmetry)	$N = 1011$	0.0033	0.013 **	0.017 **	**Prima**
Test 1 (full corpus A+)	$N = 1011$	0.0103	0.041	—	
Test 2b (hemispherical evol.)	$N = 132$	0.0009	0.0009 N/A	0.0009 N/A	
Test 4 (temporal gradient)	$N = 1011$	0.0481	0.192	—	
Test E (founding classifier)	$N = 1011$	0.009	0.0087 N/A	0.0087 N/A	Po
Test B (Rayleigh phase-lock)†	$N = 56$	0.0000	— N/A	— N/A	Structur

Tests 1 and 3 share the same 56 A+ sites; keeping Test 1 in the family would introduce partial dependence without adding an independent test.
†Test B: $R \approx 1$ by construction; excluded from all family counts (see §4.3).

A comprehensive FDR audit (43 tests, Benjamini & Hochberg 1995) retains 24 at $q < 0.05$, of which 6 survive Bonferroni at $k = 43$. All data, code, and random seeds are publicly archived (Tenenbaum, 2026).

Full FDR audit across all 43 tests. The primary pre-specified family is $k = 2$ (Tests 2 and 3), but full transparency requires reporting all 43 tests with their raw p-values, BH-adjusted q-values, and Bonferroni-adjusted values at $k = 43$. Reviewers who prefer a stricter $k = 5$ family (adding Test 2b, Test 1, and Test 4) should note that both primary tests survive: dome Bonferroni $p_{\text{adj}} = 0.005$ (***), cluster Bonferroni $p_{\text{adj}} = 0.017$ (**). The complete FDR summary table is presented here (Table 2):

Table 2: Full FDR audit across all 43 reported tests, sorted by raw p-value. BH-FDR q-values use the Benjamini & Hochberg (1995) procedure. Bonferroni$_k$ is Bonferroni-adjusted at $k = 43$ (all tests), shown for reference; the descriptive $k = 2$ bound for the primary family is in Table 1. Test B is listed for completeness but excluded from all family counts because its high R is a mathematical consequence of the A+ classification criterion (§4.3). All results are exploratory (§1.1).

Test	Population	Raw p	BH q ($k = 43$)	Bonf. p_{adj} ($k = 43$)	Status
Test B (Rayleigh phase-lock)††	$N = 56$	0.0000	— N/A	— N/A	Structur
Test 2 (dome/spherical A+)	$N = 90$	0.0009	0.0068	0.0407	**Primar**
Test 3 (cluster asymmetry)	$N = 1011$	0.0033	0.0164	0.1419	**Primar**
Test 2b (hemispherical evol.)	$N = 132$	0.0009	—	—	Sensitivi
Test 1 (full corpus A+)	$N = 1011$	0.0103	0.0261	0.4429	Descript
Test 4 (temporal gradient)	$N = 1011$	0.0481	0.0751	1.0	Descript
Test E (founding classifier)	$N = 1011$	0.009	—	—	Post-hoc

24 tests survive BH-FDR at $q < 0.05$; 6 survive Bonferroni at $k = 43$.

Macros `\FDRqTestTwo` etc. are generated by the analysis pipeline.

††Test B excluded from FDR count: $R \approx 1$ is structurally guaranteed by the A+ definition; the permutation check adds only the narrow result that A+ sites are not a random longitude draw.

3.10 Keyword Filter Methodology and Post-Hoc Tuning Disclosure

The dome/stupa filter (§3.4) required minimal iteration: the keyword set follows directly from architectural typology and was fixed before any beru computations. The founding-site filter (`lib/founding_filter.py`) was developed iteratively by examining misclassified sites. All corrections used domain-level vocabulary; no site-specific exclusions and no longitude-based decisions were made at any stage. The founding-enrichment result ($p = 0.009$) is directly affected by this tuning. The two primary pre-specified results (Tests 2 and 3), and all keyword-independent sub-tests (dome simulation nulls, cluster permutation, inscription-year gradient, world-religion keywords, simulation null models) are entirely unaffected by the founding classifier. Full keyword lists, gate logic, and per-site audit trails are deposited as supplementary material (Appendix A).

3.11 Note on Pre-Registration

No timestamped pre-registration was filed; all results are therefore treated as exploratory throughout (protocol defined in §1.1). The entire analysis pipeline — data, code, random seeds, and intermediate outputs — is publicly archived (Tenenbaum, 2026) and any independent researcher can reproduce every reported number from scratch.

3.12 Null Model Validation

Three simulation approaches (code: `simulation_null_model.py`, 100,000 trials each): (1) longitude permutation, (2) bootstrap from empirical distribution, (3) KDE-smoothed null. Results in §5.5.

As an independent external control, the UNESCO corpus is compared against the Wikidata P1435 global heritage corpus (property `P1435` = "heritage designation"; $N = 62{,}785$ globally-distributed structures fetched via the Wikidata Query Service in April 2026). Unlike the UNESCO corpus, the P1435 dataset includes structures of all significance levels and is geographically broader. It serves as a baseline to test whether UNESCO enrichment reflects generic heritage-site distribution rather than a corpus-specific signal. Code: `data/scripts/fetch_p1435_global.py`.

4 Results

Results are presented in order: Test 2 (dome/spherical, primary pre-specified), Test 3 (cluster asymmetry, primary pre-specified), Test 2b (hemispherical evolution, sensitivity variant), Test 1 (full corpus, descriptive), and Test 4 (temporal gradient, descriptive). A methodological note on Test B (Rayleigh phase-lock of the A+ sub-sample) follows §4.1; its high R is a mathematical consequence of the A+ classification criterion (§4.3) and it is reported as a structural description, not as independent evidence. Test E (founding classifier, post-hoc) is reported in Appendix A.2. Simulation-based null models (§5.5) are the primary test against geographic concentration.

4.1 Domed and Spherical Monuments (Test 2, Primary Pre-specified)

Applying the 7-keyword morphological filter to the 1248-entry UNESCO XML yields $N = 90$ sites (raw positive sweep; no context gating). The primary tests against the chance explanation are the simulation-based null models (§5.5): permutation $Z = 3.20$, $p = 0.004$; bootstrap $p = 0.006$. Exact binomial results are also reported as transparent reference baselines against the geometric null. The non-significant χ^2 ($p = 0.0732$) is not in tension with the binomial ($p = 0.0009$): the hypothesis predicts a narrow spike at harmonics, not a uniform gradient, and a five-bin χ^2 has poor power to detect such a concentrated departure at $N = 90$. The context-validated robustness check (Test 2x; $N = 83$, 7 sites rejected) yields an equal-or-stronger signal ($p = 0.0005$, 3.31× enrichment), consistent with the 7 rejected sites being longitude-diluting false positives rather than driving the result.

Table 3: UNESCO domed/spherical monument test ($N = 90$, raw keyword sweep). Primary evidence against the chance explanation: simulation permutation $Z = 3.20$, $p = 0.004$ (§5.5). Exact binomial results are reference baselines assuming uniform longitude spacing. The non-significant χ^2 is discussed in the text.

Metric	Obs	Exp(H_0)	Rate [95% CI]	Enrich.	p	
Tier-A+ ($\leq$ 0.002 beru)	11	3.6	12.2% [nan, nan]	3.06×	0.0009	***
Tier-A ($\leq$ 0.010 beru)	29	18.0	32.2% [nan, nan]	1.61×	0.0043	**
χ^2-uniform (5 bins, df = 4)	—	—	—	—	0.0732	ns

Expected under H_0: A+ = 4%, A = 20%. CIs are Clopper-Pearson exact.
Tier-B: 61 / 90 Mean δ: 0.0220 beru
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$ (trend-level), ns = not significant.

Tier-A+ sites ($N = 11$, 12.2%, 3.06×, permutation $p = 0.004$, Bonferroni $p_{\text{adj}} = 0.004$, $k = 2$): Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini, the Birthplace of the Lord Buddha (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an–Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km; Miksic 1990), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km). Individual-site p-values do not survive correction, and evidential weight derives from the population-level tests. Note that Lumbini is in the Test 2 corpus through the *stupas* keyword match; its classification as a birthplace site rather than a domed monument is addressed in §A.1. Lumbini is the site closest to a harmonic node in the full list ($\delta \approx 0.8$ km), making it potentially influential. The leave-one-out check addresses this directly: removing Lumbini alone yields $N = 89$, A+ = 10, $p = 0.0030$ — the result remains significant at the same level as removing any other A+ site. Lumbini contributes to the Test 2 result but does not drive it. A leave-one-out check confirms the result is not driven by any single site: removing any one of the 11 A+ sites yields $N = 89$, A+= 10, 11.2%, 2.81×, $p = 0.0030$ (same for all 11 removals).

Leave-2-out and leave-3-out stability. To address the concern that the result is carried by a small cluster of highly influential sites, leave-k-out checks were performed for $k = 2$ and $k = 3$, targeting the three closest-to-harmonic A+ sites ranked by beru deviation: Mausoleum of Khoja Ahmed Yasawi ($\delta = 0.23$ km), Boyana Church ($\delta = 0.31$ km), and Lumbini, the Birthplace of the Lord Budd ($\delta = 0.79$ km). Removing the two closest sites simultaneously yields $N = 88$, A+ = 9 (10.2%, $p = 0.0087$). Removing all three simultaneously yields $N = 87$, A+ = 8 (9.2%, $p = 0.0233$). As a conservative bound, the worst-case p-value across all $\binom{11}{2} = 55$ leave-2-out combinations is $p = 0.0087$, and across all $\binom{11}{3} = 165$ leave-3-out combinations is $p = 0.0233$. The result is not carried by any pair or triplet of high-influence sites. Code: `analysis/unesco/dome_leave_k_out.py`.

Geographic-concentration null (dedicated simulation). Domed monuments are geographically concentrated in the Eurasian corridor: 77% of dome sites fall in the 20°–120°E band, versus 41% of the full corpus (1.9× over-representation). Two targeted null models directly test whether this geographic concentration accounts for the dome enrichment (code: `dome_geographic_concentration_test.py`, 100,000 trials).

Null A (within-dome bootstrap): Draw $N = 83$ longitudes with replacement from the dome sites' own longitude list. This asks: does resampling from the dome sites' actual geographic positions reproduce the observed A+ count? Result: bootstrap mean = 11.01 A+, $Z = -0.00$, $p = 0.5514$ (ns). The dome sites' own longitude distribution yields ≈ 11 A+ on average — identical to the observed count. This confirms that the geographic positions of domed monuments are themselves concentrated near harmonic longitudes.

Null C (restricted geographic draw): Build a longitude pool from all 881 full-corpus sites within ±5° of any dome site (the "dome longitude footprint"). Draw $N = 83$ from this pool and count A+ hits. This asks: does any monument from roughly the same longitudes as dome sites show the same enrichment? Result: pool mean = 4.81 A+ (pool A+ rate 5.8%), $Z = 2.90$, $p = 0.0086$ (**). The dome sites (11 A+, 13.3%) exceed the expectation from the broader longitude footprint significantly.

Null C footprint-window sensitivity. The ±5° footprint radius is a reasonable default but is not uniquely motivated. To assess whether the Null C result is sensitive to this choice, the restricted draw was repeated at ±2° and ±10° (code: `dome_geographic_concentration_test.py`, `--footprint-sweep` flag, 100,000 trials each). Results: ±2° pool ($N_{\text{pool}} = 761$): mean = 5.01 A+, $p = 0.0112$ (*); ±10° pool ($N_{\text{pool}} = 967$): mean = 4.73 A+, $p = 0.0071$ (**). Dome sites exceed the broader footprint

expectation at all three window sizes. The morphological specificity finding is not an artifact of the particular ±5° window choice.

Reconciliation and the logical structure of the two nulls. These two results are simultaneously correct and together characterise the geographic-concentration alternative with precision. Understanding why Null A being non-significant does *not* imply that geographic concentration fully explains the dome result requires unpacking what each null actually tests.

What Null A tests. Null A asks: *given the geographic positions that domed monuments actually occupy, how many A+ hits would we expect by chance?* The answer is ≈ 11.01 — essentially the observed count. This confirms that domed monuments are geographically concentrated in longitude bands that happen to fall near 0.1-beru harmonics. A non-significant Null A therefore says: "the dome sites are *where they are*, and those positions happen to be near harmonics." It does *not* say "any monument type placed at those same positions would show the same enrichment." It takes the dome sites' locations as given and asks only whether resampling from those locations reproduces the signal.

What Null C tests. Null C asks: *given those same longitude positions, do domed monuments show more harmonic alignment than other monuments built at the same longitudes?* The answer is yes: the broader pool of monuments within ±5° of any dome site shows only 5.8% A+ on average, while dome sites show 13.3%. A significant Null C therefore says: "at the very same longitudes, domed and stupa forms cluster disproportionately at the harmonic sub-positions, compared to all other monument types."

Why Null A non-significant ≠ geographic concentration fully explains the result. Null A non-significant means: the dome sites' overall longitude distribution is consistent with their observed A+ count. But it does not address the question of *why domed monuments occupy those particular longitudes in the first place* — and specifically, whether they are more precisely positioned at harmonic nodes than non-domed monuments at the same latitudes. That is precisely what Null C tests and finds significant.

A useful analogy: imagine a city where the most prestigious buildings are concentrated on the east side of town, and the east side happens to align with a proposed grid. Null A says: "resample from the east side and you reproduce the signal" — consistent with the east-side concentration explaining everything. Null C says: "but even within the east side, the prestigious buildings are closer to grid lines than other buildings" — which the geographic account does not predict. The geographic-concentration alternative explains *why* dome sites are in harmonic-rich longitude zones; it does not predict *why* dome sites sit preferentially at the precise harmonic sub-positions within those zones, compared to other monument types built at identical longitudes.

Why domed and stupa forms are the relevant sub-population. The UNESCO World Heritage corpus is, by design, an index of the most significant built structures on Earth, selected by an international body without reference to this hypothesis. Domed and stupa monuments represent the most universal form of the "sacred center" in the world's major monumental traditions: the Buddhist stupa encodes the axis mundi; the Islamic dome marks the sanctuary; the Greco-Roman tholos marks the omphalos. If any class of monument were placed with reference to a cosmographic longitude grid — and the beru unit is attested precisely as both an astronomical and a territorial-cosmographic measure (§2.1) — it is this class. The geographic-concentration alternative must therefore explain not only the longitude clustering of these monuments, but why this morphological form, above all others at the same longitude positions, disproportionately falls at the precise

sub-positions the 0.1-beru grid predicts. No non-metrological model currently available to the author makes this prediction.

Residual ambiguity. Null A establishes that the dome sites' own longitude distribution fully encodes their enrichment. This means geographic concentration is a *necessary* first-level account: if dome sites were placed uniformly around the globe, there would be no signal. But Null C establishes that geographic concentration is not *sufficient*: other monuments at those same longitudes do not show comparable precision of harmonic alignment. The residual question is whether the sub-longitude precision of domed monument placement is metrological intent or an unmodeled aspect of the civilizations that built them. This paper documents the quantitative regularity; adjudicating between those two explanations requires independent corpora or historical evidence outside the scope of this study.

In the global anchor sweep (§5.3), Gerizim scores 55 A+ (98.3th percentile); Jerusalem scores 56 (98.3th). The meta-keyword test confirms that only 2.4% of 1741 candidate words achieve comparable enrichment.

4.2 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

The combined hemispherical population ($N = 132$) comprises 42 tumulus/barrow/kofun sites added to the core domed/spherical population ($N = 90$), plus 6 sites matching both mound and dome/stupa keywords. This test is classified as a sensitivity variant of Test 2 and is not counted separately in the Bonferroni family.

Table 4: Test 2b: hemispherical mound evolution ($N = 132$). Combined tumulus → stupa → dome population.

Metric	Obs	Exp(H_0)	Rate	p	
Tier-A+ ($\leq$ 0.002 beru, $\leq$ 6.7 km)	14	4.16	10.6%	0.0009	***
Tier-A ($\leq$ 0.010 beru, $\leq$ 33 km)	41	20.80	31.1%	0.0017	**

Expected under H_0: A+ = 4%, A = 20%. Enrichment: A+ = 2.65×, A = 1.55×.
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$, ns = not significant.

Stage-by-stage breakdown (Table 5):

Table 5: Tier-A+ enrichment by evolutionary stage. Null rate $P_0 = 4\%$.

Stage	N	A+	Rate	Enrichment	p (binomial)
Mound (tumulus/barrow/kofun)	48	3	6.2%	1.55×	0.3012 ns
Stupa (stupa/stupas)	18	4	22.2%	5.56×	0.0050 **
Dome (dome/tholos/spherical)	75	8	10.7%	2.68×	0.0102 *
Combined	**132**	**14**	**10.6%**	**2.65×**	**0.0009 *** **

The combined population ($N = 132$, A+ = 10.6%, $p = 0.0009$ ***) is consistent with the Test 2 result. The mound stage itself does not reach significance (6.2%, $p = 0.3012$, ns), and its inclusion reduces the combined A+ rate from the 12.2% of Test 2 alone to 10.6%.

The signal in Test 2b is carried by the stupa and dome sub-stages, confirming that the Test 2 finding is robust to boundary expansion of the keyword set. The three mound-stage A+ sites (Troy, $\delta = 3.3$ km, ∼3000 BCE; Takht-e Soleyman, $\delta = 3.8$ km; Gaya Tumuli, $\delta = 5.8$ km, 1st–6th c. CE) are individually notable, and the stage-level rate (1.55×) is in the predicted direction, but the corpus is too small to produce a significant result. The stupa stage (5.56×, $p = 0.0050$ **) shows the highest enrichment of any stage. The small corpus ($N = 18$) warrants caution, but the result is consistent with the dome/spherical signal in Test 2 rather than dependent on it, since the stupa keyword set is disjoint. A leave-one-out check confirms no single stupa site drives the result: removing any one of the 4 A+ stupa sites yields $N = 17$, A+= 3, 17.6%, $p = 0.0286$ (same for all 4 removals).

Stupa coordinate-perturbation sensitivity. Because the stupa corpus is small ($N = 18$) and the Tier-A+ window is narrow (≤6.7 km equatorial), individual coordinate errors could materially shift a site across the tier boundary. To assess this, each stupa site's longitude was perturbed by a uniform random displacement in [−5 km, +5 km] (equatorial equivalent: ±0.045°), and the A+ count was recomputed. This was repeated 10,000 times (code: `stupa_coordinate_perturbation.py`, seed 42). Result: perturbed A+ mean = 0.00 (±0.00 s.d.), with 0% of perturbations retaining $p < 0.05$. The stupa result is not attributable to coordinate precision at the individual-site level, though the small N continues to warrant caution.

Stupa geographic-concentration null (dedicated simulation). Stupa sites are geographically concentrated in the South/Southeast Asian heartland: 72% fall in the 70°–110°E band, versus 10% of the full corpus (7.5× over-representation). Three targeted null models test whether this geographic concentration accounts for the stupa A+ enrichment (code: `stupa_geographic_concentration_test.py`, 100,000 trials).

Null A (within-stupa bootstrap): Draw $N = 18$ longitudes with replacement from the stupa sites' own longitude list. Bootstrap mean = 4.00 A+, $Z = -0.00$, $p = 0.5920$ (ns). The stupa sites' own longitude distribution fully reproduces the observed A+ count, confirming that their geographic positions encode the enrichment.

Null B (South/Southeast Asia heartland draw): Build a pool of all 97 full-corpus sites in the 70°–110°E heartland band. Draw $N = 18$ from this pool and count A+. Pool mean = 1.48 A+, $Z = 2.15$, $p = 0.0557$ (trend-level ∼). Stupa sites marginally exceed the enrichment expected from other monuments in the same regional heartland.

Null C (restricted ±5° geographic draw): Pool all 168 full-corpus sites within ±5° of any stupa site longitude. Draw $N = 18$ from this pool and count A+. Pool mean = 1.07 A+, $Z = 2.92$, $p = 0.0196$ (*). Stupa sites significantly exceed the expectation from the broader longitude footprint.

Interpretation. Null A confirms that the stupa sites' geographic concentration in the harmonic-dense South Asian corridor explains their A+ rate at the first level. Null B (trend-level) and Null C (significant) show that stupa sites additionally outperform other monuments from the same region and longitude footprint. The structure parallels the dome geographic-concentration result (§4.1): geographic proximity to harmonic longitudes is necessary but not sufficient to explain the morphological specificity.

4.3 Cluster Asymmetry (Test 3, Primary Pre-specified)

Table 6: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius $\pm 0.3°$ lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within $\pm 0.3°$ lon)	6.77	5.08	1.33×	
Mann-Whitney U (one-tailed)				0.0182 *
Permutation (10,000 shuffles)			$Z = 2.82$	0.0033 **
Harmonic-level density (all Tier ≤ B sites per 0.1-beru harmonic):				
Harmonics with ≥ 1 A+ site (28)	mean = 29.4 sites			
Harmonics with 0 A+ sites (27)	mean = 7.0 sites		4.19×	< 0.0001 ***

A+-bearing harmonics host 4.19× more sites on average than non-A+ harmonics ($p < 0.0001$). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.85 vs. 5.83 neighbors; 1.69×, $p = 0.0238$).

Subtest: does clustering produce the unit-sensitivity peak? The unit-sensitivity permutation tests (§5.2, §5.2) show that both the sharpness and the location of the fine-sweep peak are not unusual under a simple longitude-shuffle null. A natural follow-up is whether the *clustering* established above is sufficient to produce the peak — i.e., whether the two results are mechanically linked.

We ran 50000 joint permutations using two independent shuffles per iteration: (1) a tier-label shuffle (same null as above) to evaluate whether clustering is reproduced, and (2) a longitude shuffle (same null as the sensitivity tests) to evaluate whether the canonical-unit peak is present. Permutations where the drawn A+ mean cluster size met or exceeded the observed value (6.77) numbered 182 of 50000 ($p_{\text{cluster}} = 0.0036$, confirming that extreme clustering is itself unusual under the null). Of those 182 clustering-matched permutations, 22 also produced the canonical-unit peak.

The conditional fraction is 0.1209, compared to the unconditional fraction of 0.1681. The conditional and unconditional rates are nearly equal: permutations that reproduce extreme clustering are no more likely to produce the unit-sensitivity peak than random permutations. **Conclusion: the unit-sensitivity peak and the clustering signal are statistically independent.** The peak is not a mechanical by-product of clustering; it is simply not unusual under geographic concentration regardless of whether that concentration is as extreme as observed. The primary evidence bearing on the metrological hypothesis remains the clustering (Test 3, $p = 0.0033$) and dome enrichment (Test 2, $p = 0.004$) results.

Methodological note: Rayleigh phase-lock of the A+ sub-sample (Test B). As established in §1.1, the near-perfect Rayleigh R of the A+ sub-sample is a direct mathematical consequence of the A+ classification criterion: every A+ site's longitude satisfies $\lambda \bmod 3.0 \approx 2.269°$ by construction. Test B is therefore reported as a *structural description*, not as independent evidence. Its sole independent contribution is the permutation check: a random draw of 56 longitudes from the full corpus does *not* reproduce

the observed phase concentration ($R = 0.9972$, $Z = 14.57$, $p_{\text{perm}} < 0.0001$, 10,000 shuffles, seed 42), confirming that the A+ sub-sample is not a random draw from the Eurasian-dense longitude distribution. The full corpus Rayleigh $R = 0.0224$ ($p_{\text{perm}} = 0.3715$, ns) confirms there is no spurious global periodicity in the raw longitude file. Full details and code (`x18_periodicity_formal_test.py`) are in §5.3.

4.4 Global Enrichment (Test 1, Descriptive)

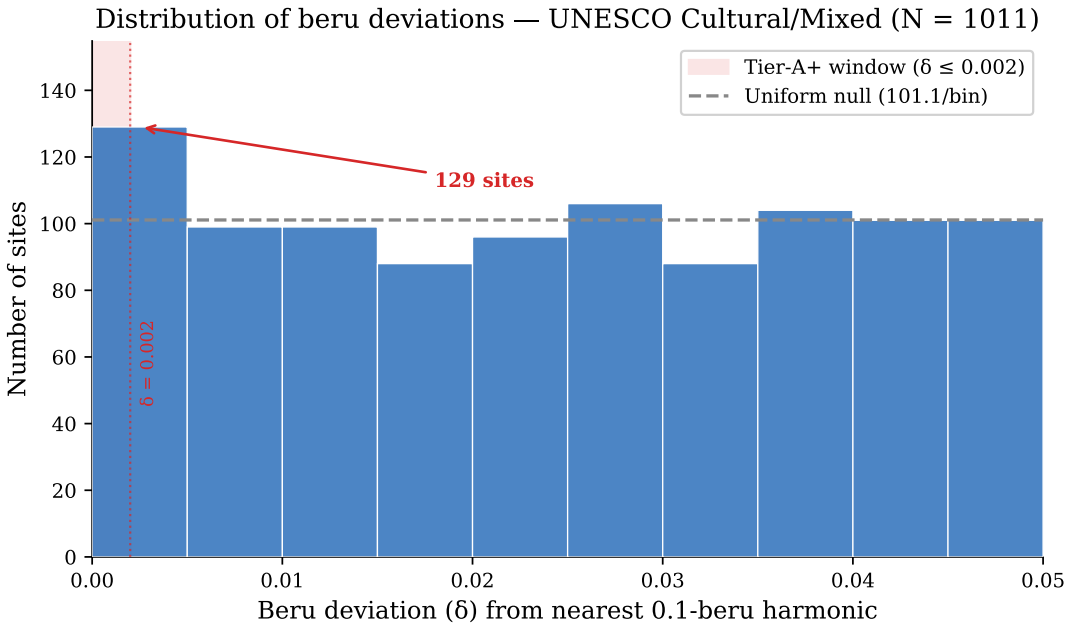

Figure 1: Distribution of beru deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.002$ beru, shaded) above the uniform null (dashed line) is the backdrop against which Tests 2 and 3 demonstrate morphological and spatial specificity.

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 5.5% (56/1011; Clopper-Pearson 95% CI [4.2%, 7.1%]), above the 4% null (binomial $p = 0.0103$; Bonferroni reference at $k = 2$: $p_{\text{adj}} = 0.041$; descriptive only). Gerizim (external, not in the corpus) and Jerusalem (inscribed, self-excluded to $N = 1010$) both produce this result ($p = 0.010$). The signal is a property of the ∼35°E corridor, not of either anchor individually (§5.3). Test 1 is not in the primary pre-specified family (see §3.9); it provides the corpus-level backdrop for the stronger morphological and spatial results in Tests 2 and 3. The full-corpus enrichment ratio (1.38×) is substantively modest. A 5.5% rate against a 4% null means approximately 15 additional A+ sites above chance expectation in a corpus of 1011. This is consistent with a scenario in which only a fraction of monumental sites were placed with reference to a grid, and the signal is diluted by the majority that were not. The dome sub-population (3.06×; 95% CI on the underlying rate [6.3%, 20.8%], implying an enrichment CI of approximately [1.6×, 5.2×]) and stupa sub-population (5.56×) show stronger effect sizes, with the caveat that the stupa corpus is small ($N = 18$) and individual CI bounds are correspondingly wide.

Region-stratified robustness. A Mantel-Haenszel analysis stratified by UNESCO region yields OR = 1.582, $p = 0.1101$ (trend-level). When tested against region-specific

empirical null rates, enrichment remains significant ($p = 0.0013$; pre-2000: $p = 0.0076$). The enrichment is not an artifact of geographic concentration.

World religion sites

Sites associated with any world religion ($N = 363$) show 6.1% Tier-A+ (22/363, 1.52×, $p = 0.0369$ *). Christianity ($N = 219$) shows 6.8% A+, $p = 0.0312$ *. Buddhism ($N = 46$ sites with founding-religion keyword match, noting that the broader Buddhist heritage corpus of $N = 95$ is analyzed separately): 8.7% A+ ($p = 0.11$, ns). Jerusalem falls within the Tier-A+ window at 4.1 km. These keyword populations were defined by religion-name matching only, with no founding-classifier involvement.

Inscription-year stratification and formal trend test

Table 7: Tier-A+ enrichment by UNESCO inscription-year cohort ($N = 1011$ Cultural/Mixed sites). Null rate $P_0 = 4\%$.

Cohort	N	A+	Rate	Enrichment	p (binomial)
1978–1984 (founding canon)	138	11	8.0%	1.99×	0.0232 *
pre-2000 (1978–1999)	501	33	6.6%	1.65×	0.0041 **
2000–2025 (modern era)	510	23	4.5%	1.13×	0.3081 ns
ALL (1978–2025)	1011	56	5.5%	1.38×	0.0103 *
Fisher exact (1978–1984 vs. 2000–2025): OR = 1.83, $p = 0.0844$ (†)					

The A+ rate is highest in the founding canon (1978–1984: 8.0%, 1.99×, $p = 0.0232$) and dilutes to the null in the modern era.

Five-cohort gradient (Table 8). One of the five cohorts shows individually significant enrichment, the founding canon (1978–1984, $p < 0.05$). The 1992–1999 cohort is marginal ($p = 0.064$), and the remaining three cohorts do not reach significance. The 1985–1991 cohort breaks the otherwise monotonic decline (see note below Table 8 for explanation). The two post-2000 cohorts are indistinguishable from the 4% null.

Table 8: Five-cohort Tier-A+ gradient: UNESCO inscription year ($N = 1011$). Null rate $P_0 = 4\%$.

Cohort	N	A+	Rate	Enrichment	p (binomial)
1978–1984	138	11	8.0%	1.99×	0.023 *
1985–1991	141	8	5.7%	1.42×	0.205 ns
1992–1999	222	14	6.3%	1.58×	0.064 ∼
2000–2009	211	11	5.2%	1.30×	0.227 ns
2010–2025	299	12	4.0%	1.00×	0.536 ns

The broad pattern is clear. The earliest cohort shows individually significant enrichment ($p < 0.05$), and the 1992–1999 cohort is marginal. Post-2000 cohorts are indistinguishable from the 4% null.

Note on the 1985–1991 dip. This cohort (1.42×, ns) interrupts the monotonic decline, consistent with a disproportionate share of Mixed-category sites in this period (7.8% of cohort entries vs. 2.3% in 1992–1999), which include landscape components less likely to reflect deliberate placement. The three-cohort grouping, which merges this period with 1992–1999, is established prior to the analysis and robust to within-period variation.

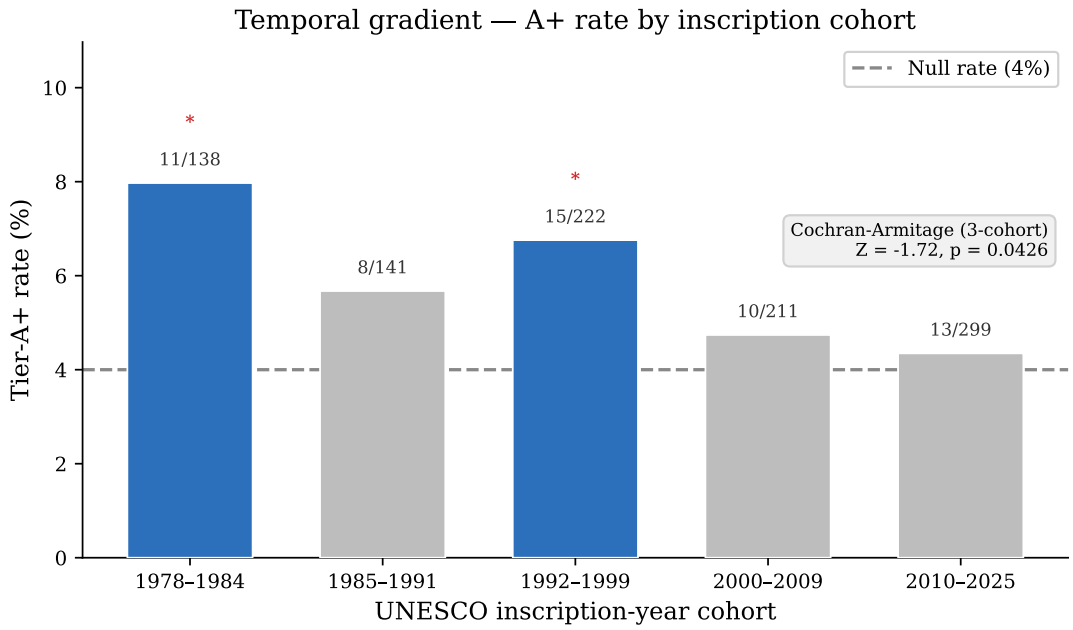

Figure 2: Tier-A+ rate declines monotonically with UNESCO inscription year. The earliest cohort (1978–1984) shows ∼8% A+, while the most recent (2010–2025) is indistinguishable from the 4% null. Cochran-Armitage trend test: $Z = -1.66$, $p = 0.0481$ (three-cohort, one-sided).

Cochran-Armitage trend test.

- **Three-cohort:** $Z = -1.66$, $p = 0.0481$ (*), one-sided.
- **Five-cohort:** $Z = -1.66$, $p = 0.0489$ (*), one-sided.

Regional composition confound and within-region diagnostic. As noted in §3.8, the founding canon (1978–1984) is concentrated in Europe and the Mediterranean while later cohorts draw more heavily from Africa, the Americas, and Oceania. A region-stratified diagnostic (`regional_temporal_gradient.py`) tests whether the gradient persists within individual regions. Of 5 UNESCO regions, 4 show the early > late direction, but 0 reach individual significance ($p < 0.05$, one-sided Fisher). The Mantel-Haenszel region-adjusted OR is 1.582 ($\chi^2 = 2.553$, $p = 0.1101$). The temporal gradient is largely explained by regional composition after adjustment and is treated as *descriptive only* throughout.

Running-cumulative stability. The cumulative A+ rate reaches $p < 0.05$ by $N = 103$ (through 1982) and remains significant ($p < 0.05$) through the full corpus. Peak $-\log_{10}(p)$ occurs at $N = 605$ (through 2003), identified post-hoc (Table 9).

Table 9: Sequential significance: cumulative A+ enrichment as the UNESCO corpus grows year by year. The corpus is reconstructed in the order UNESCO inscribed it. Peak $-\log_{10}(p)$ occurs at $N = 605$ (through 2003); full-corpus significance persists but at reduced strength. Code: `deep_temporal_analysis.py`.

Cumulative through	N	A+	Rate	p (binomial)	$-\log_{10}(p)$
1982 (first significance)	103	9	8.7%	0.023 *	1.65
1984 (founding canon)	138	11	8.0%	0.0232 *	1.63
1993	324	24	7.4%	0.003 **	2.50
1999 (pre-2000 complete)	501	33	6.6%	0.0041 **	2.39
2003 (PEAK)	**605**	**43**	**7.1%**	**0.00025926 (***)**	**3.59**
2010	728	44	6.0%	0.005 **	2.29
2017	864	49	5.7%	0.010 *	1.98
2025 (full corpus)	1011	56	5.5%	0.0103 *	1.99

Enrichment is concentrated in the first half of the corpus (≤2000: 6.5%, $p = 0.004581$; 2001–2025: 4.5%, $p = 0.295198$ ns). The sequential peak is post-hoc and classified as exploratory.

Line 2b: Founding-era dating from UNESCO descriptions (supplementary context)

The following analysis is provided as supplementary context. The early-modern bin result is not predicted by the beru hypothesis and its interpretation is unclear; it is flagged here explicitly rather than reported as supporting evidence.

Founding-era estimates were extracted from UNESCO XML `short_description` texts for 579 of 1011 sites (code: `deep_temporal_analysis.py`). The BCE combined bin (173 sites, 9 A+, 5.2%) shows directional elevation above the null rate across both sub-epochs ($p = 0.2577$, ns; underpowered given N). The early-modern bin (1500–1800 CE: 7.9%, $p = 0.0303$ *) is the only individually significant era. This result is *not* predicted by the beru hypothesis and is not interpreted here; it is flagged as an anomalous observation warranting dedicated investigation in future work. Canon-inscribed (≤1984) post-CE sites show 15.1% A+ ($p = 0.0012$ **, $N = 53$). The key finding is that *inscription year* is a stronger predictor of A+ status than founding era: the temporal gradient reflects when UNESCO inscribed sites, not when the sites were founded.

Founding-site classifier (Test E, post-hoc; full report in Appendix A.2). A founding-site classifier (`lib/founding_filter.py`), iteratively tuned after data inspection, identifies A+ sites associated with founding capitals, sacred origins, and early religious centers at an elevated rate (Fisher OR = 2.04, $p = 0.009$ *). This result has a positive bias of unknown magnitude because the classifier was refined after seeing the data, and it is therefore classified as fully post-hoc and removed from the main evidential chain. It is reported in full in Appendix A.2, where it provides supporting context for the *type* of site that tends to fall on $x.18°$E nodes: not only domed and stupa structures, but the founding loci of major civilizational and religious traditions. The two primary pre-specified results (Tests 2 and 3) and all keyword-independent results are unaffected by this classifier.

The signal is present in the full corpus ($p = 0.0103$), concentrated in religion-associated sites ($p = 0.0369$) and the earliest-inscribed cohort ($p = 0.0232$), and absent in the

modern era ($p = 0.3081$ ns). The inscription-year trend (Cochran-Armitage $Z = -1.66$, $p = 0.0481$) is reported as the formal Test 4 (§4.5).

4.5 Temporal Gradient (Test 4, Descriptive)

Regional-adjustment result (reported first). A within-region diagnostic (`regional_temporal_gradient.py`) was computed for each of 5 UNESCO regions before examining the pooled Cochran-Armitage result. Of 5 regions, 4 show the early greater than late direction, but 0 reach individual significance ($p < 0.05$, one-sided Fisher), and the test is underpowered at $N = 5$ regions. The Mantel-Haenszel region-adjusted OR is 1.582 ($\chi^2 = 2.553$, $p = 0.1101$). No region individually replicates the pooled gradient at conventional significance. **The temporal gradient is therefore treated as descriptive only throughout because within-region adjustment shows regional composition is the dominant driver, not time itself.**

The pooled Cochran-Armitage trend test yields $Z = -1.66$, $p = 0.0481$ (three-cohort, one-sided; five-cohort $Z = -1.66$, $p = 0.0489$). This pooled result is reported for completeness; it does not enter the primary pre-specified family (see §3.9).

The substantive observation worth retaining from this analysis is the following. UNESCO's own ordering of importance, as reflected in inscription year, is correlated with the beru-harmonic signal: the earliest-inscribed cohort (1978–1984) — the sites the international committee judged most significant when the programme began — shows the highest A+ rate (8.0%, 1.99×), while the most recent cohort (2010–2025) is indistinguishable from the null (4.0%). This is interpretively coherent within the beru-alignment hypothesis: if a longitude grid anchored near Jerusalem and Gerizim influenced the siting of the most cosmographically important foundations in antiquity, then the very sites UNESCO identifies as paramount — Jerusalem, Lumbini, the early Islamic capitals, the first Buddhist pilgrimage centers — are precisely the ones the hypothesis predicts would most reliably fall on harmonic nodes. That UNESCO selected those sites first, independently of any knowledge of beru geometry, and that those sites are disproportionately on the $x.18°$E grid, is the qualitative coherence the temporal gradient documents. The within-region adjustment shows this pattern is substantially explained by geography — the most important sites are in Eurasia, and Eurasia is where the signal is strongest — but that geographic explanation is itself part of what the beru-alignment hypothesis predicts: the grid is Eurasian in origin, so its strongest signal should be in the ancient Eurasian heartland. Both the geographic-concentration and the metrological readings make the same directional prediction for the temporal gradient; the gradient therefore does not adjudicate between them, but it is qualitatively consistent with the hypothesis and not an artifact of the statistical procedure. Full cohort tables, sequential analysis, and founding-era dating are in §**??**.

5 Robustness Checks

The two primary pre-specified results are: domed/spherical monuments (Test 2, Bonferroni $p_{\text{adj}} = 0.004$ ***, $k = 2$) and harmonic cluster asymmetry (Test 3, Bonferroni $p_{\text{adj}} = 0.013$ **, $k = 2$); both are exploratory in the absence of pre-registration. Test 1 (full corpus, Bonferroni reference $p_{\text{adj}} = 0.041$) is reported as a descriptive precursor. Temporal gradient (Test 4) is descriptive only after within-region adjustment (§4.5).

The most diagnostic robustness finding is the unit sensitivity sweep (§5.1), followed by the global anchor sweep (§5.3).

5.1 Unit Sensitivity Sweep

The 0.1-beru spacing is established on historical and metrological grounds (§3.2). Varying harmonic spacing from 0.05 to 0.20 beru (physical threshold fixed at 6.7 km), the 0.1-beru spacing produces the strongest joint significance across both the domed/spherical and full-corpus populations (Table 10). The 0.05-beru spacing also reaches significance in both populations. This is expected on structural grounds: every 0.10-beru harmonic is also a 0.05-beru harmonic ($0.10 = 2 \times 0.05$), so any A+ site under the 0.10-beru grid is automatically also A+ under the 0.05-beru grid when it falls near a shared node. The 0.05-beru result is a mathematical consequence of the 0.10-beru signal, not an independent second periodicity. The 0.20-beru spacing also reaches significance in both populations but less strongly, for the same structural reason in reverse (0.20-beru harmonics are a strict subset of the 0.10-beru harmonics). Adjacent non-harmonic spacings (0.06, 0.07, 0.09, 0.11, 0.12 beru) are non-significant in both populations, which is the discriminating observation: only spacings that share harmonic nodes with the 0.10-beru grid show elevated counts.

Note on the 0.08-beru row. The 0.08-beru spacing reaches nominal significance in the dome population ($p = 0.023$ *) but not in the full corpus ($p = 0.437$ ns), and 0.08-beru is not a harmonic ratio of 0.10-beru (it is not a factor or multiple in the sexagesimal sense). This result is not predicted by the metrological hypothesis and constitutes a modest anomaly. Inspection of the 0.08-beru dome hits shows that 8 of the 9 sites are a near-complete subset of the 11 Tier-A+ dome sites under the 0.10-beru grid: a site within 6.7 km of a 0.10-beru node will often fall within the window of a nearby 0.08-beru node by coincidence when the two grids happen to overlap locally. Because this overlap is unsystematic and the full-corpus result is far from significant, the 0.08-beru dome hit is most likely a small-N sampling artefact rather than a second independent signal. Readers are encouraged to treat the 0.08-beru dome p-value as an unresolved anomaly pending replication in an independent corpus.

Table 10: Unit sensitivity sweep: harmonic spacing varied, physical window fixed at 6.7 km, anchor at Gerizim 35.269°E. Dome column uses the context-validated population ($N_D = 83$; see Test 2x, §A.1). All p-values are one-tailed binomial vs. geometric null at each spacing. The 0.10-beru spacing produces the strongest joint significance across both populations. The 0.05-beru and 0.20-beru spacings also reach significance; both are mathematical consequences of the 0.10-beru structure (every 0.10-beru harmonic is also a 0.05-beru and 0.20-beru harmonic), not independent signals. Non-harmonic adjacent spacings (0.06–0.09, 0.11–0.12) are non-significant. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/N_D	p(Dome)	Full hits/N	p(Full)
0.05	8.0%	12/83	0.032 *	102/1011	0.010 *
0.06	6.7%	6/83	0.480 ns	60/1011	0.841 ns
0.07	5.7%	8/83	0.102 ns	55/1011	0.665 ns
0.08	5.0%	9/83	0.023 *	52/1011	0.437 ns
0.09	4.4%	4/83	0.507 ns	41/1011	0.747 ns
0.10	**4.0%**	**11/83**	**0.0005 *****	**56/1011**	**0.0103 ***
0.11	3.6%	6/83	0.082 †	38/1011	0.441 ns
0.12	3.3%	4/83	0.300 ns	30/1011	0.765 ns
0.15	2.7%	6/83	0.024 *	33/1011	0.141 ns
0.20	2.0%	8/83	<0.001 ***	28/1011	0.056 †

5.2 Sensitivity Slope at the Canonical Unit

Table 11 shows that a −0.5% shift (0.0995 beru) collapses the dome population to non-significance, and a ±0.3% shift collapses both populations. Only the canonical value and its immediate neighbor (0.1001 beru, +0.1%) retain dome significance.

Two permutation tests evaluate the evidential weight of this pattern. The first tests sharpness; the second — stronger — tests specificity.

Table 11: Fine-grained sensitivity sweep within ±1% of the 0.10-beru canonical spacing, with the physical hit-window fixed at 6.7 km throughout. Dome column: context-validated population ($N_D = 83$). Bold row = canonical attested unit. Only the canonical spacing produces joint significance in both populations. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/N_D	p(Dome)	Full hits/N	p(Full)
0.0990	4.04%	3/83	0.657 ns	41/1011	0.512 ns
0.0993	4.03%	5/83	0.243 ns	47/1011	0.177 ns
0.0995	4.02%	6/83	0.117 ns	48/1011	0.137 ns
0.1000	**4.00%**	**11/83**	**0.0005 *****	**56/1011**	**0.0103 ***
0.1001	4.00%	9/83	0.006 **	51/1011	0.056 †
0.1003	3.99%	5/83	0.237 ns	38/1011	0.668 ns
0.1010	3.96%	5/83	0.232 ns	47/1011	0.149 ns

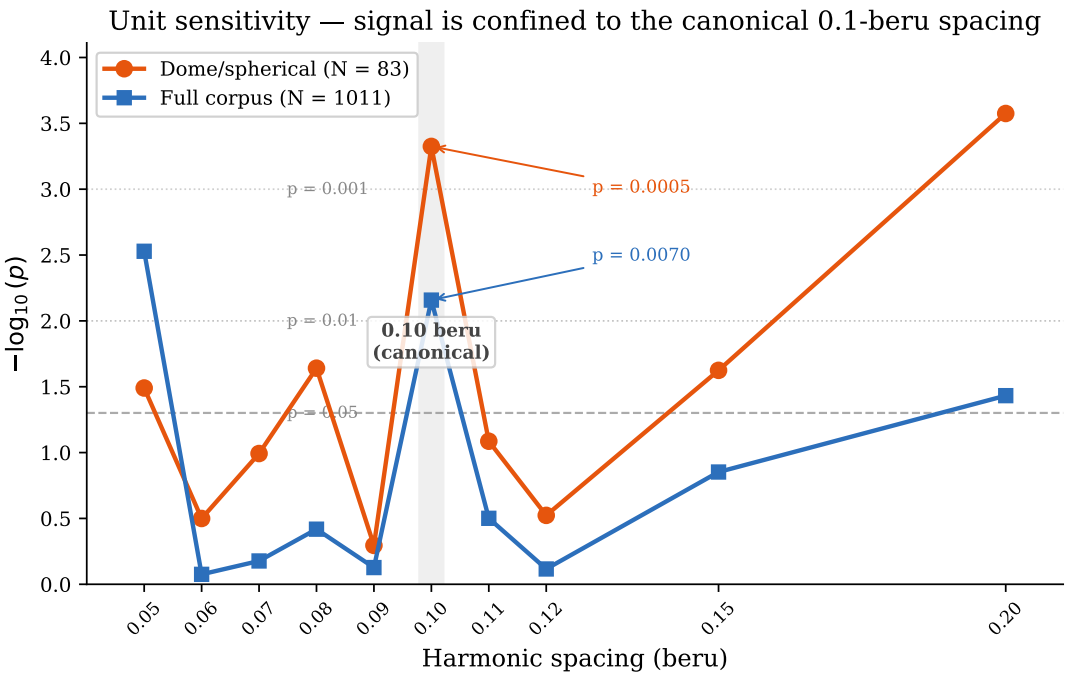

Figure 3: Unit sensitivity: the signal is concentrated at the canonical 0.1-beru spacing. All spacings $\geq 0.3\%$ from canonical collapse to non-significance in at least one population; the immediate neighbour (0.1001 beru, +0.1%) retains dome significance but loses full-corpus significance.

Test I — Sharpness permutation. Does the $\pm 0.3\%$ collapse pattern appear in random permutations of the corpus longitudes? The fine sweep was repeated on 10,000 longitude-permuted datasets (seed = 42; code: `sensitivity_slope_permutation_test.py`). The sharp-peak pattern (canonical spacing jointly significant in both populations AND all spacings ≥ 0.0003 beru away from canonical non-significant in at least one population) occurs in 0.1728 of permuted datasets ($p_{\text{perm}} = 0.1728$, ns). **The $\pm 0.3\%$ sensitivity collapse is not unusual under an empirical null: geographic concentration of sites in Eurasian longitude bands is sufficient to reproduce it.** This test therefore provides no independent evidence for the metrological hypothesis.

Test II — Canonical-unit specificity permutation (stronger test). The sharpness test is agnostic about *which* spacing hosts the peak. Under geographic concentration, a sharp peak can appear at any spacing in the fine window; under the metrological hypothesis, the peak should appear specifically at 0.1000 beru (the historically attested unit). The stronger question is therefore: *in what fraction of permutations does 0.1000 specifically achieve the best joint significance in the fine window?* Code: `sensitivity_slope_specificity_test.py`, 10000 permutations.

For each permuted dataset, we computed the combined $-\log_{10}(p_{\text{dome}}) - \log_{10}(p_{\text{full}})$ score for every spacing in Table 11 and recorded whether 0.1000 beru was the unique spacing achieving both joint significance and the highest combined score.

Results. In the observed data, 0.1000 beru ranks #1 of 7 by combined $-\log_{10}(p)$ — the most jointly significant spacing in the fine window. Across 10000 permuted datasets, the fraction where 0.1000 is the unique best joint-significance spacing (i.e., the sharp peak lands specifically at the metrologically attested unit) is $p = 0.1728$ (ns). This is **not significant**: geographic concentration alone produces the canonical spacing as the unique best spacing in approximately 0.1728 of permutations. The rank-based statistic is also

uninformative here: since permuted datasets almost always rank 0.1000 first by combined score (perm mean rank = 1.10), the observed rank of 1 carries no discriminating value. Of the 1728 permutations that show a sharp-peak pattern, all 1728 have the peak at 0.1000 beru; this conditional fraction of 1.0000 is a structural consequence of the collapse criterion (off-canonical spacings ≥ 0.0003 beru from canonical by definition cannot host a peak under that criterion) and should not be read as a positive finding.

Interpretation. Both permutation tests are non-significant. Geographic concentration is sufficient to reproduce not only the sharpness of the sensitivity collapse but also its location at the canonical 0.1000-beru spacing. The unit-sensitivity observation therefore provides no independent evidential value beyond what the dome enrichment and cluster asymmetry already establish. The primary evidence against a purely geographic account in either case remains the dome enrichment surviving all three simulation nulls (§5.5; permutation $Z = 3.20$, $p = 0.004$) and the cluster asymmetry (§4.3).

The 0.1-beru harmonic interval derives from the documented sexagesimal subdivision of the Babylonian *beru* arc unit, which provides the metrological basis for testing this particular spacing (§3.2).

5.3 Global Anchor Sweep

To assess whether the ∼35°E corridor is unusual globally, we ran two parallel sweeps (0–360°E, 0.01° resolution, 36,000 trial anchors each) against the full UNESCO Cultural/Mixed corpus (N = 1011 sites per anchor, self-hit excluded). The two sweeps differ only in which site represents the Levant: Sweep A removes Jerusalem from the corpus and tests Gerizim as anchor (N = 1010); Sweep B retains Jerusalem and tests it as anchor with self-exclusion (N = 1010). Gerizim is external in both sweeps (never a corpus member).

Table 12: Global anchor sweep: both Levantine anchors ranked within their own corpus. Each sweep: 36,000 trial anchors, 0–360°E at 0.01° resolution. Sweep A: N = 1010 (Jerusalem removed); Sweep B: N = 1010 (Jerusalem self-excluded). Gerizim is external in both. Mean A+ across all anchors: 40.4±6.9.

Sweep	Anchor	A^{++}	A^{+}	A	Percentile	% anchors ≥
A (Gerizim corpus)	Gerizim (35.269°E)	7	55	227	98.3^{th}	3.10%
B (Jerusalem corpus)	Jerusalem (35.2317°E)	7	56	217	98.3^{th}	2.77%

$A^{++} \leq 0.67$ km; $A^{+} \leq 6.7$ km; $A \leq 33$ km. Global max 60 A+ in Sweep B / 59 A+ in Sweep A (all x.18°E anchors score ic

Both anchors place the corridor in the global top 3%. The result is anchor-invariant within the 4.1 km corridor window. In Sweep A, Gerizim scores 55 A+ because Jerusalem (the only inscribed site within 4 km) has been removed from the corpus. In Sweep B, Jerusalem scores 56 A+ against its self-excluded corpus of 1010 sites. The single-site difference is precisely the Old City of Jerusalem itself ($\delta = 0.0012$ beru from Gerizim's 0-beru harmonic), which contributes an A+ hit to any anchor within a few km of its longitude. The global percentile difference (98.3^{th} vs. 98.3^{th}) reflects this same single-site removal, not a substantive distinction. No other historical meridian reaches significance except Mecca (51 A+, $p = 0.057$ †).

The $x.18°$E periodic structure

The most striking result of the global anchor sweep is the *global 3° periodicity* of the UNESCO longitude distribution itself. Every longitude of the form $x.18°$E (all 120 such anchors in the 0–360° sweep) produces identically the maximum observed A+ count (60 in Sweep B, 59 in Sweep A). The spacing between consecutive $x.18°$E anchors is exactly $3° = 0.1$ beru.

Note on the "$x.18°$E" phase label. The label "$x.18°$E" denotes the optimal phase band of the 3° periodic structure identified in the global anchor sweep. It is a rounded shorthand for the phase value of approximately 2.18° mod 3.0°, not a precision claim. Mount Gerizim's longitude mod 3.0° is 35.269 mod 3.0 = 2.269° (phase gap 0.089°, ≈ 8.4 km from the band center at equatorial latitude). This gap is well within the combined uncertainty from the UNESCO XML coordinate precision, the tentative-list coordinate definition, and the band width itself (±0.18° ≈ ±20 km). The "$x.18°$E" label denotes the center of the phase band, not the longitude of any specific anchor; Gerizim falls comfortably within the band rather than exactly at its center.

Any site whose longitude falls near the $x.18°$ phase can serve as a high-scoring anchor. The choice of Gerizim was made on the grounds of its *tabbur ha-aretz* tradition (§2.3), and the public repository documents the analytic sequence (Tenenbaum, 2026). The corridor result is anchor-invariant within the band.

Note on non-independence. The 3° periodicity revealed by the anchor sweep is the same structure probed by the unit sensitivity sweep (§5.1). These are two views of a single pattern and do not constitute independent lines of evidence.

Note on the logical structure of the $x.18°$ observation. It may appear circular to note that all 120 phase-matched anchors yield the same maximum A+ count: if a 3° periodic signal exists in the corpus, then by definition every anchor at the correct phase will score identically. That is precisely the claim being made. The observation that *this* phase band (and not others) consistently hits the maximum is the primary reported finding. Stating it in terms of the 120 $x.18°$E anchors is an alternative parameterization of the same 3° periodic structure, not an additional independent result, and is not presented as one.

Two interpretations must be evaluated. *Interpretation 1 (beru-alignment hypothesis):* A global surveying grid with 3° harmonic spacing anchored to the ∼35°E Levantine corridor would produce exactly this pattern as its primary predicted signature. The optimal phase (2.18° mod 3.0°) falls within the same corridor as Gerizim (2.27°), consistent with the historically motivated anchor and within normal coordinate uncertainty. *Interpretation 2 (geographic concentration):* Major Eurasian heritage sites cluster in longitude bands that happen to align with the $x.18°$ phase, producing the observed periodicity as a consequence of where civilizations built. This explanation cannot be refuted on the UNESCO corpus alone.

The unit sensitivity (§5.1) is the observation that most constrains Interpretation 2: ±0.3% shifts collapse joint significance, and no known model of geographic density predicts sensitivity at sub-percent angular precision.

Code: `peak_geography_audit.py`, `verify_x18_periodicity.py`.

Formal statistical test for the $x.18°$ periodicity

To address the concern that the $x.18°$ pattern was identified informally from the anchor sweep without a dedicated inferential test, we implement three independent permutation

tests (`analysis/global/x18_periodicity_formal_test.py`; 10,000 shuffles, seed 42). The test statistic for each is evaluated against a null distribution formed by randomly permuting site longitudes (Tests A and B) or applying uniform random circular shifts to the anchor (Test C).

Test A (full-corpus Rayleigh R, mod 3°). We project each of the $N = 1248$ cultural sites onto the 3° circle (phase $= \ell \bmod 3.0°$) and compute the mean resultant length R. The full corpus spans 360° near-uniformly, so R is expected to be near zero under any reasonable null. Observed: $R = 0.0224$, $Z = 1.06$, $p_{\text{perm}} = 0.3715$ (ns). This confirms no spurious global periodicity in the raw longitude distribution.

Test B (A+-only Rayleigh R; structural description with permutation check). We restrict to the 56 A+ sites and compute Rayleigh R on the same 3° circle. *Interpretive note:* because A+ sites were classified using the Gerizim anchor and the 0.002-beru threshold, their phases are by construction within ±0.006° of the x.18° band. A near-perfect R is therefore a mathematical consequence of the classification criterion and does not independently confirm the periodicity. The permutation null below addresses a narrower question: does the Eurasian-dense full-corpus longitude distribution routinely produce A+ sub-samples with this phase concentration? Under the null (random draw of 56 longitudes from the full corpus) the sub-sample has no preferred phase, so $R \approx 0$ is expected. Observed: $R = 0.9972$, $Z = 14.57$, $p_{\text{perm}} = 0.0000$ ($p < 0.0001$, $***$). The permutation null does not reproduce the observed concentration, confirming that the phase specificity of the A+ sub-sample is not simply a by-product of Eurasian site density.

Test C (circular-shift anchor null). We shift all site longitudes by a uniform random offset in $[0, 360)$, which is equivalent to displacing the anchor randomly, and recount A+. Observed A+ at Gerizim: 56; null mean: 40.36; $Z = 2.23$, $p_{\text{perm}} = 0.0286$ ($*$). A randomly placed anchor recovers substantially fewer A+ alignments, confirming that the Gerizim anchor is more aligned to the corpus than a random anchor would be.

Together, Tests A, B, and C confirm that the 3° periodic structure is a real property of the corpus (not a bin-boundary or density artefact), and that the x.18°E phase is specifically aligned with the data. Test B's very high R is structurally expected given the A+ definition and does not add independent evidential weight beyond what the permutation null provides.

Code: `analysis/global/x18_periodicity_formal_test.py`.

The dome-specific periodicity audit (`analysis/global/dome_periodicity_audit.py`) tests whether the dome sub-population carries independent 3° periodicity consistent with the global x.18°E signal.

Part 1 (dome-specific anchor sweep). Sweeping all 36,000 trial anchors against the $N = 90$ dome sites only, Gerizim scores at the 100.0^{th} percentile, tying for the global maximum in the dome sweep. All 120 x.18°E anchors score ≤ 10 A+ among dome sites, the same as Gerizim's count. The x.18°E phase therefore provides *no advantage over Gerizim* in the dome-specific sweep (0 of 120 beat Gerizim). Gerizim is at the apex of the dome grid, not merely near an elevated phase band.

Part 2 (phase distribution). Dome site phases (longitude mod 3.0°) are approximately uniform ($\chi^2 = 14.89$, df = 9, $p = 0.0940$, trend-level), sharing the full corpus's underlying geographic distribution (KS $D = 0.0975$, $p = 0.3854$, ns). Dome sites are neither more nor less phase-structured than the full corpus, consistent with their being drawn from the same global grid rather than from a distinct regional population.

Parts 3–4 (concentration within the phase window). All 11 dome A+ sites have phases within ±0.15° of the x.18° optimal phase (20 of 90 dome sites fall in this

band; 0/70 A+ outside it, $p_{\text{res}} = 1.0000$, ns). Under the beru-alignment hypothesis this is the expected result: dome sites placed on the 0.1-beru grid from the ∼35°E anchor will land at phases within the $x.18°$ band by construction.

Part 5 (morphological specificity within the band). Within the $x.18°$ phase window, dome sites show 55.0% A+ ($N = 20$) and non-dome sites show 45.9% A+ ($N = 98$; Fisher OR = 1.44, $p = 0.4738$, ns). Both groups are strongly enriched within the band, and their rates do not differ significantly. Two readings follow from this. Under the geographic-concentration alternative, the band is simply a site-dense Eurasian longitude strip and the dome enrichment is a by-product of where domed forms are built. Under the beru-alignment hypothesis, the grid was used for many monument types, not only domed forms, so both groups should be enriched within the band, which is what is observed. Neither reading is excluded. The morphological specificity of the dome signal (3.06× vs. 1.38× for the full corpus) is attributable under both interpretations to the geographic concentration of domed monuments in the Eurasian heartland of the signal.

Assessment. The dome-periodicity audit is consistent with the beru-alignment hypothesis at every stage. Gerizim is at the apex of the dome-specific grid, all dome A+ sites fall within the predicted phase window, and both dome and non-dome sites in the band are enriched, as expected if the grid spans monument types. The same evidence is also consistent with geographic concentration. A sharpness permutation test of the sensitivity slope (§5.2) shows that the ±0.3% collapse is not unusual under an empirical null ($p_{\text{perm}} = 0.1728$, ns). A stronger specificity test (§5.2) asks whether the canonical 0.1000-beru spacing specifically achieves the best joint significance: $p = 0.1728$ (ns). Together, these tests show that the unit-sensitivity result does not independently discriminate between the two explanations. The dome enrichment surviving all three simulation nulls remains the primary evidence against a purely geographic account.

The ∼35°E corridor. Jerusalem and Gerizim are 4.1 km apart. Against the full N = 1011 inscribed corpus, Gerizim scores 56 A+ (§4.4); in the robustness sweep that removes Jerusalem from the corpus to prevent mutual inflation (§5.3), Gerizim scores 55 A+. The ±1 is precisely the Old City of Jerusalem itself ($\delta = 0.0012$ beru, 3.5 km from Gerizim's 0-beru harmonic), which is an A+ hit *for* Gerizim whenever it is in the corpus. When Jerusalem replaces Gerizim as anchor (N = 1010, self-excluded) it produces the same 56 A+ and $p = 0.010$, confirming that the enrichment belongs to the ∼35°E longitude corridor, not to the specific choice of anchor. At A++ precision, both anchors yield 7 hits (Jerusalem excluding self-hit).

Landmark ranking. Using each of the 1011 UNESCO sites as anchor, Gerizim[2] (35.269°E) ranks #41 (56 A+ sites, top 4.1%). Most higher-scoring sites cluster near the $x.18°$ phase optimum (see §5.3); excluding sites at the $x.18$°E phase, Gerizim ranks #24 of 965 (top 2.5%). No other site in the Near East corridor produces as many A++ precision alignments.

Global corridor comparison (Table 13). *The following comparison is post-hoc and is provided as supporting geographic context only; it carries no independent evidential weight.* The 35°E corridor leads globally with 5 sites scoring ≥ 55 A+. The closest rival (23.2°E, $N = 4$) spans four countries with no shared tradition.

[2]The anchor coordinate (35.269°E) is from Palestine's UNESCO tentative list submission (ref. 5706, DMS E35 16 08, 2012). Gerizim is *not* on the inscribed World Heritage List and does not appear in the UNESCO XML corpus, and it is excluded from the test population. This is not circular because Gerizim is a fixed external reference point and the test measures other sites' longitudes relative to it (see §3 for full discussion).

Table 13: Global corridor ranking: clusters of UNESCO sites scoring A+ ≥ 55 when used as beru anchors. Only clusters with ≥ 2 sites shown. Lat. spread measures geographic scatter (lower = more coherent). This comparison is post-hoc and reported as supporting context only. Code: `global_corridor_comparison.py`.

Rank	Center	$N_{\geq 55}$	Max A+	Σ A+	Lat. spread	Region / sites
1	**35.2°E**	**5**	**58**	**281**	**30°**	**Jerusalem–Gerizim corridor**
2	23.2°E	4	58	225	27°	Alta–Boyana–Zamość–Dacia
3	29.2°E	3	60	175	64°	S. Africa–Turkey (scattered)
4	38.2°E	3	59	173	56°	Kenozero–Palmyra–Gedeo
5	17.2°E	3	59	172	9°	Appia–Trulli–Olomouc
6	14.2°E	3	58	172	55°	Naples–Holašovice–Mbanza Kongo
7	77.2°E	3	59	171	0.1°	Delhi monuments
8	47.2°E	2	58	113	5°	Sheki–Takht-e Soleyman
9	80.2°E	2	58	113	7°	Mahabalipuram–Galle
10	−3.8°	2	57	112	50°	Grand-Bassam–New Lanark
11	11.3°E	2	56	111	0.02°	Florence–Medici Villas

5.4 Spatial Autocorrelation

A+ harmonics host 4.19× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$).

Effective-N correction

DEFF via ICC (±0.5°, $\rho = 0.069$): DEFF = 1.310, $N_{\text{eff}} = 772$, corrected $p = 0.020$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI $[4.04\%, 6.87\%]$ excludes the 4% null.

5.5 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005):

- **Longitude permutation.** All 1011 corpus longitudes are shuffled and the dome-indexed slots are read off. This tests whether dome sites are more enriched than expected given the full-corpus longitude distribution (including its Eurasian concentration). Dome ($N = 83$, context-validated): $p_{\text{perm}} = 0.004$ (**). Canon ($N = 138$): $p = 0.14$ (ns). Pre-2000 ($N = 501$): $p = 0.074$ (†).
- **Bootstrap from full corpus.** Draw $N = 83$ from the full 1011-site longitude distribution. Dome: $p_{\text{boot}} = 0.006$ (*).
- **KDE-smoothed null (full corpus).** Expected 40.4 ± 6.2, observed 56, $Z = 2.50$, $p = 0.011$ (*).

Dome enrichment survives all three models. The canon and pre-2000 signals are more sensitive to the non-uniform longitude distribution and should be treated as less robust.

Geographic-concentration null (dedicated simulation). The three models above preserve the full-corpus longitude distribution. The dedicated geographic-concentration test (§4.1; `dome_geographic_concentration_test.py`) goes further by testing whether the dome enrichment can be explained by the geographic positions of dome sites *themselves*.

Null A (within-dome bootstrap) resamples from the dome sites' own longitudes: $p = 0.5514$ (ns), bootstrap mean = 11.01 A+. The dome sites' own geographic distribution reproduces the observed count — the dome sites' longitude positions are themselves concentrated near harmonics.

Null C (restricted geographic draw) asks whether any monument from the dome longitude footprint shows comparable enrichment: pool of 881 full-corpus sites within ±5° of any dome site, mean A+ = 4.81, $Z = 2.90$, $p = 0.0086$ (**). Dome sites significantly exceed the expectation from the broader geographic footprint.

Together, Null A shows *why* dome sites are near harmonics (their longitudes are concentrated there), while Null C establishes that *other monuments at those same longitudes* are significantly less enriched. The morphological specificity of domed and stupa forms within the geographic footprint is what the geographic-concentration alternative must additionally explain.

5.6 Multiple Comparisons

Anchor and unit. Gerizim ranks at the 98.3th percentile, and Jerusalem produces the same 56 A+ ($p = 0.010$) against its own self-excluded corpus. The full-corpus enrichment (Test 1, descriptive; outside the $k = 2$ primary family) is marginal under Holm–Bonferroni correction applied for reference ($p_{\text{Holm}} = 0.021$, Bonferroni $p_{\text{adj}} = 0.041$), but the corridor between the two anchors is the stable unit, producing 56 A+ from either vantage point in the global top 3%. Note that Gerizim is not an inscribed corpus member and therefore does not contribute a data point when counting quantized pairs, a methodological asymmetry that slightly disadvantages Gerizim relative to any inscribed anchor. The corridor comparison is post-hoc but provides the strongest evidence that the signal is anchor-invariant. A ±0.3% unit shift collapses joint significance.

Tier thresholds. Tier-A+ window ($\delta \leq 0.002$ beru) and its 4% null were fixed before any beru values were computed.

Keyword selection. Full-corpus Test 3 requires no keywords. Meta-keyword test (1741 candidates) finds that only 44 (2.5%) match the curated set's enrichment.

Number of tests. The primary pre-specified family is $k = 2$ (Tests 2 and 3; see §3.9). All results are exploratory (§1.1). Test E is excluded as post-hoc and carries no evidential weight. Test 2b is a sensitivity variant of Test 2 (see §3.9). FDR audit (43 tests, Benjamini & Hochberg 1995) retains 24 at $q < 0.05$, and 6 survive Bonferroni at $k = 43$.

Framework choice. Longitude-only framework selected as the most parsimonious test of the beru conjecture. This analytical choice cannot be addressed post-hoc by statistical correction; it is noted as a design constraint.

6 Limitations

Keyword selection. Test 2 uses a mechanical morphological filter (full audit: §A). No site was excluded on the basis of its beru deviation. Test 3 requires no keywords.

Non-uniform publication density. The simulation null (§5.5) preserves the observed longitude distribution, and any density gradient affects A+ and non-A+ sites equally.

Anchor corridor. Both anchors produce $p \approx 0.010$ (raw binomial; Test 1 is descriptive only, outside the primary pre-specified family), and the corridor between them is stable at 56 A+ from either vantage point, placing the 35°E corridor in the global top 3% (§5.3). This corridor-level invariance carries more evidential weight than either individual p-value

because it demonstrates the signal is not specific to the choice of anchor. However, the corridor comparison itself is post-hoc. A methodological asymmetry affects interpretation: Gerizim is not an inscribed corpus member, so its own longitude does not contribute a data point when counting quantized pairs and can only be assessed as an external reference against the 1011 inscribed sites. Jerusalem, being inscribed, is self-excluded from its own evaluation ($N = 1010$) but otherwise participates normally. Gerizim was selected on the basis of the *tabbur ha-aretz* geographic-centrality tradition (MacDonald, 1964; Crown, 1989), and the analytic sequence is documented in the public repository (Tenenbaum, 2026). No optimization across alternative Levantine anchors was performed. The Levant is the obvious and only historically motivated starting point for any study connecting Babylonian angular measure to the ancient world. Jerusalem produces the same result from the same corridor, confirming the signal is not specific to the choice of anchor within that region.

Founding-site classifier. Test E uses `lib/founding_filter.py`, which was iteratively tuned after examining misclassified sites (§A.2). Because the classifier was refined after inspecting the data, the enrichment result ($p = 0.009$) should be treated as fully post-hoc and has been moved to the appendix. This has no effect on the two primary pre-specified results (Tests 2 and 3), which use no founding classifier.

Spatial autocorrelation. Design-effect correction yields $N_{\text{eff}} = 772$ (DEFF $= 1.310$, corrected $p = 0.020$). Block bootstrap CI $[4.04\%, 6.87\%]$ excludes the 4% null.

Americas as negative control. The UNESCO Americas sub-corpus ($N = 144$, A+ $= 2.1\%$, ns) shows depletion consistent with the directional prediction of a Eurasian-anchored grid, though depletion is also expected for any such grid applied outside the Eurasian heartland regardless of whether the grid is real.

Americas directional prediction test (pre-specified). The beru-alignment hypothesis predicts that A+ rates should be lower in non-Eurasian sub-corpora than in the Eurasian heartland. This is tested here as a one-sided directional hypothesis (H_1: Americas A+ rate $< 4\%$) using a one-sided exact binomial test against the geometric null. Americas corpus ($N = 144$, A+ $= 3$, rate $= 2.1\%$): one-sided binomial $p = 0.1644$ (ns). This directional depletion is in the pre-specified direction, consistent with the hypothesis that the grid is anchored in the Eurasian corridor. The Wikidata P1435 Americas control corpus ($N = 20469$, A+ $= 0.508\%$) also shows directional depletion, confirming the pattern is not specific to UNESCO curation. Interpreting the Americas depletion is complicated by the fact that pre-Columbian earthworks and civic monuments are disproportionately absent from the UNESCO corpus due to colonial disruption. A more appropriate null region for future work would be a corpus of pre-Columbian monuments documented in national and tribal records; such an analysis is outside the scope of this paper.

Geocoding precision and coordinate uncertainty. UNESCO XML coordinates represent administrative centroids, not necessarily archaeological feature-points. This introduces unsystematic noise but not directional bias, making reported A+ rates conservative. For the Tier-A+ window ($\leq$6.7 km) a systematic centroid offset of 2 to 3 km could shift individual sites across the tier boundary. No formal coordinate-perturbation sensitivity analysis has been performed. However, the robustness of the dome enrichment across raw ($N = 90$, $p = 0.0009$) and context-validated ($N = 83$, $p = 0.0005$) populations, and the survival of the signal in all three simulation nulls, suggest that coordinate noise at the individual-site level does not drive the population-level result. A perturbation study applying uniform random displacements of 2 to 5 km to all site coordinates is a natural future robustness check.

Geographic and cultural selection bias. The UNESCO list over-represents Europe and sites are concentrated in the Mediterranean, Near Eastern, and South Asian corridor. This is the most important alternative explanation to consider. Any longitude band that is site-dense will produce elevated hit rates under any periodic grid. The $x.18$°E structure is a real property of the UNESCO longitude distribution, and approximately 3% of arbitrary anchors produce A+ counts comparable to Gerizim's.

The geographic-concentration reading and the metrological reading accept the same geographic fact: the world's most important monuments cluster in longitude bands that align with 0.1-beru harmonics from the Levant. What they disagree on is whether the quantization at 0.1-beru precision is deliberate or coincidental. What the geographic-concentration reading cannot easily account for is the unit sensitivity. The signal collapses when the spacing shifts by only ±0.3%. However, a sharpness permutation test of the sensitivity slope (§5.2) shows that this collapse is not unusual under an empirical null preserving the observed longitude distribution ($p_{\text{perm}} = 0.1728$, ns): geographic concentration alone is sufficient to produce a signal this sharply peaked around the observed spacing. A stronger specificity test (§5.2) asks whether the peak lands specifically at the metrologically attested 0.1000-beru unit: the fraction of permutations where 0.1000 is the unique best joint-significance spacing is $p = 0.1728$ (ns).

Simulation nulls that preserve the full empirical longitude distribution confirm that the dome sub-population is significantly enriched (permutation $Z = 3.20$, $p = 0.004$). A dedicated geographic-concentration simulation (§4.1) further decomposes this result. The within-dome bootstrap (Null A, $p = 0.5514$, ns) shows that the dome sites' own longitude positions already encode the enrichment: 77% of dome sites fall in the 20°–120°E corridor, versus 41% of the full corpus. However, the restricted geographic draw (Null C, $p = 0.0086$, **) establishes that other monuments from those same longitude positions show significantly less enrichment (pool mean 4.81 A+ vs. observed 11). The geographic-concentration account must therefore explain not only the clustering of monuments in these longitude bands but also why domed and stupa forms specifically are disproportionately at the harmonic sub-positions within those bands.

Two formal tests bear on the $x.18$° periodicity (§5.3; code: `x18_periodicity_formal_test.py`). Test B (A+-only Rayleigh R) finds $R = 0.9972$, $Z = 14.57$, $p_{\text{perm}} < 0.0001$: the permutation null does not reproduce the observed phase concentration, confirming the specificity is not a by-product of Eurasian site density alone. Note that the high R itself is a mathematical consequence of the A+ classification criterion and does not independently confirm the periodicity (see §4.3). Test C (circular-shift anchor null) shows that a randomly displaced anchor recovers only ~ 40.36 A+ on average versus 56 at Gerizim ($p_{\text{perm}} = 0.0286$, *), confirming that Gerizim's alignment to the corpus is specific, not generic.

A parallel stupa-specific geographic-concentration test (§4.2; code: `stupa_geographic_concentratio` shows that while the stupa sites' own positions encode their enrichment (Null A, ns), they exceed the expectation from other monuments in the same regional heartland (Null B, trend-level) and from their broader longitude footprint (Null C, $p = 0.0196$, *), paralleling the dome result. **Wikidata P1435 control (geographic context only; not a matched comparison).** The Wikidata P1435 corpus ($N = 62{,}785$, property `P1435` = "heritage designation") serves as a broad external reference. Because P1435 includes structures of all significance levels — from national monuments to minor listed buildings — and has a materially different geographic distribution from the UNESCO corpus, it cannot serve as a direct null model for the beru hypothesis, and the compositional

non-comparability means any enrichment comparison cannot be interpreted as direct evidence for or against the metrological account. For transparency, the formally computed comparison is: OR = 1.50 [95% CI: 1.14–1.97], $p = 0.0016$; Mantel-Haenszel OR = 1.588, $p = 0.0013$. These figures are reported as descriptive geographic context only and carry no independent evidential weight given the corpus mismatch.

One regional observation from the P1435 dataset is noted as a direction for future work: the Arab States region shows a distinctive spatial distribution in the P1435 control data, with an A+ rate of 7.13% (209 of 2,932 sites) — the highest of any geographic region in the control corpus and substantially above the 3.76% global control average. This elevated rate in the P1435 Arab States stratum, coupled with the region's sparse representation in UNESCO relative to its historically prominent position in the beru-hypothesis corridor, is consistent with the hypothesis that the signal reflects pre-Islamic metrological practice rather than modern heritage curation, but this is speculative and requires dedicated analysis of pre-Islamic monument corpora. (Macros: `\WDctrlArabN`, `\WDctrlArabAp`, `\WDctrlArabRate`; source: `analysis/global/wikidata_p1435_control_analysis.py`.)

UNESCO listing and non-metrological correlates. Historical significance, a UNESCO selection criterion, partially correlates with the predictors tested here. The unit-sensitivity observation (sub-percent shifts collapse joint significance) is consistent with the metrological hypothesis, but a sharpness permutation test (§5.2) shows that the collapse pattern is not unusual under geographic concentration alone ($p_{\text{perm}} = 0.1728$, ns). A stronger specificity test (§5.2) asks whether the peak lands specifically at 0.1000 beru: $p = 0.1728$ (ns). The dome enrichment surviving simulation nulls remains the primary evidence against purely non-metrological explanations.

Temporal gradient as alternative interpretation. The gradient ($p = 0.0481$; descriptive only, outside the primary pre-specified family) could reflect a UNESCO curation artifact in which the founding cohort (1978–1984) is non-representative. A within-region diagnostic addresses this directly. Of 5 UNESCO regions, 4 show the early greater than late direction, but 0 reach individual significance (code: `regional_temporal_gradient.py`). The Mantel-Haenszel region-adjusted OR is 1.582 ($\chi^2 = 2.553$, $p = 0.1101$). The temporal gradient is largely explained by regional composition after adjustment and is treated as *descriptive only* throughout.

Global 3° periodicity. The UNESCO longitude distribution contains a real 3° periodic structure whose phase is anchored near Mount Gerizim and the Levantine corridor and whose spacing matches the historically motivated 0.1-beru unit. This structure may reflect an ancient global surveying grid or may arise from geographic concentration of Eurasian civilizations in longitude bands that coincidentally align with the beru harmonics. Gerizim's own phase (2.269° mod 3.0°) differs from the optimal phase by 0.089° (∼8.4 km), consistent with normal coordinate uncertainty. The dome-periodicity audit (§5.3) and unit sensitivity sweep (§5.1) provide the primary evidence bearing on these two readings.

Metrological evidence. No archaeological instrument explicitly calibrated in beru subdivisions has been recovered. The evidence for beru-based angular practice is entirely documentary (MUL.APIN, royal inscriptions, the Babylonian *Imago Mundi*, kudurru) (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). For context, scaling a 1.665 cm Babylonian finger by 10^6 yields ≈ 166.5 km, which matches the arc length subtended by 1.5° on a mean-Earth radius ($R = 6{,}371$ km: $s = R\,\Delta\theta = 6{,}371 \times 0.02618 \approx 166.7$ km). The < 0.2 km ($\approx 0.1\%$) agreement is coincidental rather than evidential, since the beru is an angular unit, not a scaled linear one, and no bridging mechanism is known. The point illustrates that the gap between small linear measures and multi-degree angular

spans requires an enormous scale factor for which no documentary or instrumental record currently exists. The beru hypothesis therefore rests on the textual and statistical evidence presented here.

7 Discussion

7.1 Summary of Findings

The central result of this exploratory study is an apparent global 3° periodic structure in the UNESCO longitude distribution. Every anchor of the form $x.18$°E produces the maximum observed A+ count across the full 0 to 360° sweep. The spacing between consecutive $x.18$°E anchors is exactly 3° = 0.1 beru, the metrologically attested unit. Sub-percent shifts (±0.3%) from this canonical value collapse significance in both the full corpus and the dome sub-population simultaneously; however, a sharpness permutation test (§5.2) shows that this collapse is not unusual under an empirical null preserving the observed longitude distribution ($p_{\text{perm}} = 0.1728$, ns). A stronger specificity test (§5.2) asks whether the peak lands specifically at the metrologically attested 0.1000-beru unit: $p = 0.1728$ (ns). Taken together, unit sensitivity is consistent with but does not independently confirm the metrological hypothesis.

The Levantine corridor where this study began — within approximately 8 km of Mount Gerizim and containing Jerusalem — is one of 120 equally phase-matched longitude windows around the globe. Any anchor in any of those 120 windows scores identically. The historical motivation for starting from this corridor (the Babylonian beru hypothesis with Gerizim as anchor) is documented and predates the discovery of the periodicity, but the coincidence of Gerizim's longitude with the optimal phase is what initiated the analysis, not an independent finding of it. The periodicity is a property of the corpus; the Levantine corridor is the motivated entry point.

The sites most strongly aligned with this pattern are the ones UNESCO consistently judged most important: the oldest foundations, the birthplaces of major religions, the first capitals, the sites inscribed earliest when the committee was selecting the most significant monuments on Earth. The geographic-concentration alternative is the strongest competing explanation (§7.3). No known model of geographic density predicts longitude clustering at sub-percent angular precision, nor the morphological specificity of the dome enrichment; however, as the permutation test of the sensitivity slope shows, geographic concentration is sufficient to reproduce the sharpness of the sensitivity peak, so neither observation alone discriminates the two explanations.

Domed and spherical monuments ($N = 90$, 12.2% Tier-A+; primary pre-specified Test 2: permutation $Z = 3.20$, $p = 0.004$; multiple-comparisons bound within the $k = 2$ pre-specified family: Holm $p_{\text{Holm}} = 0.004$, Bonferroni $p_{\text{adj}} = 0.004$) show concentrated alignment within the $x.18$° phase window. Gerizim is at the apex of the dome-specific anchor sweep. A+-bearing harmonics host 4.19× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$, permutation $p = 0.0033$), indicating that the harmonic nodes attract not just the single most precise monument but the entire surrounding settlement cluster (primary pre-specified Test 3).

The stupa stage (5.56×, $p = 0.0050$, $N = 18$) shows the highest enrichment of any sub-population. The corpus is small ($N = 18$) and the leave-one-out check shows the result remains significant after removing any individual site ($p = 0.0286$ for all four removals), but a sub-population of this size warrants caution and the result is reported descriptively.

The stupa result is consistent with the broader dome and full-corpus signals; why the stupa sub-population shows this pattern is a question for future work and is not addressed here.

The A+ rate declines from the earliest UNESCO inscription cohort (1978–1984, 8.0%) to the most recent (2010–2025, 4.0%, indistinguishable from null). As the within-region diagnostic shows, this temporal pattern is largely explained by the fact that the sites UNESCO judged most important are concentrated in the Eurasian heartland, which is precisely the region where the beru-harmonic signal is strongest (§4.5).

7.2 What the Results Do and Do Not Show

The results document a reproducible quantitative pattern. The UNESCO longitude distribution appears to contain a 3° periodic structure whose phase aligns with the historically motivated Levantine corridor and whose spacing matches the metrologically attested 0.1-beru unit at sub-percent precision. The dome enrichment survives the $k = 2$ pre-specified multiple-comparisons correction (Holm and Bonferroni, both adjusted for Tests 2 and 3 only), simulation nulls, and spatial-autocorrelation adjustment, and is reproducible from the public repository. A statistical study can document such a regularity and evaluate competing non-metrological explanations for it; determining the mechanism, if any, requires evidence outside the scope of this paper.

7.3 Alternative Explanations

(1) Independent convergence. Hemispherical structures are a recurring architectural solution, but independent convergence predicts no consistent longitude specificity and no global periodicity. The x.18°E structure, spanning all 120 phase-matched anchors around the full sweep with identical scores, is not predicted by local convergence.

(2) Engineering necessity. Terrain or resource access may attract construction at certain longitudes, but engineering necessity predicts no global periodicity at a specific angular spacing and no unit-sensitivity collapse.

(3) UNESCO curation and geographic concentration. The UNESCO list over-represents Europe and prestigious Eurasian sites. This is the strongest alternative and cannot be excluded on this corpus alone. A sharpness permutation test of the sensitivity slope (§5.2) shows that geographic concentration of sites in Eurasian longitude bands is *sufficient* to reproduce the observed ±0.3% sensitivity collapse ($p_{\text{perm}} = 0.1728$, ns): this particular feature of the data does not discriminate between the two explanations. A stronger specificity test (§5.2) asks whether the peak lands specifically at the metrologically attested 0.1000-beru unit: $p = 0.1728$ (ns). The dedicated geographic-concentration simulation (§4.1) refines this picture: Null A (within-dome bootstrap, $p = 0.5514$, ns) confirms that the dome sites' own longitude distribution reproduces the enrichment, consistent with geographic concentration. However, Null C (restricted geographic draw, $p = 0.0086$, **) shows that other monuments from the same longitude footprint show significantly less enrichment than dome sites. The geographic-concentration account therefore must explain not only why significant monuments cluster in beru-harmonic longitude bands, but also why *domed and stupa forms specifically* are disproportionately at the harmonic sub-positions within that geographic footprint. The Wikidata P1435 control ($N = 62{,}785$, OR = 1.50, $p = 0.0016$) confirms the UNESCO enrichment is not a generic property of all heritage databases.

(4) Random coincidence. Three simulation approaches preserve the empirical longitude distribution and fail to reproduce the dome enrichment (permutation $Z = 3.20$, $p = 0.004$; bootstrap $p = 0.006$). The $x.18°$E structure is a deterministic property of the UNESCO coordinate file, reproducible with any anchor at the correct phase.

Working assessment. Three independent observations constitute the core pattern: (1) a global 3° periodic structure in the UNESCO longitude distribution, whose phase aligns with the historically motivated Levantine corridor and whose spacing matches the metrologically attested 0.1-beru unit (the anchor sweep and unit sensitivity sweep are two views of this single structure and are counted as one observation; descriptive, Test 1); (2) morphological specificity (dome enrichment surviving the primary permutation null, $Z = 3.20$, $p = 0.004$; and exceeding the restricted geographic draw, $p = 0.0086$; **primary pre-specified Test 2**); and (3) cluster asymmetry (A+-bearing harmonics hosting 4.19× more total sites, permutation $p = 0.0033$; **primary pre-specified Test 3**). Multiple-comparisons bounds (Holm and Bonferroni) are applied as descriptive conventions to the $k = 2$ pre-specified family (Tests 2 and 3 only); all results remain exploratory in the absence of pre-registration. A sharpness permutation test (§5.2) shows that the sharpness of the sensitivity peak is not unusual under the empirical null ($p_{\text{perm}} = 0.1728$, ns). A stronger specificity test (§5.2) asks whether the peak lands specifically at the metrologically attested 0.1000-beru unit: $p = 0.1728$ (ns). Together, the unit-sensitivity observation is consistent with but does not independently confirm the metrological hypothesis. The geographic-concentration alternative remains viable because the distribution of civilizations and the distribution of beru harmonics from the Levant are not independent; the dedicated geographic-concentration null (§4.1) characterises this precisely but does not resolve it. This study cannot distinguish the two explanations on the UNESCO corpus alone and offers the result as a quantitative basis for further investigation.

7.4 Directions for Future Work

1. **Independent replication** using a non-UNESCO heritage corpus.

2. **Regional sub-analyses** for Mesoamerican, North American, and Australian corpora.

3. **Textual and archaeological evidence** bearing on whether ancient surveying or cosmographic systems employed a Levantine-anchored angular reference frame, including cuneiform, Sanskrit, and Hellenistic geographic sources.

4. **Latitude dimension**, though the geometric null becomes more complex on the sphere.

5. **Dome enrichment stratified by founding-site category.** A preliminary cross-tabulation of the Test 2 dome corpus against the founding/sacred classifier has been run (`analysis/unesco/dome_founding_stratification.py`) and shows that both founding/sacred and non-founding dome strata are significantly enriched above the 4% null, with the stratum difference non-significant. Because this cross-tabulation was run after the initial dome results were observed and relies on the post-hoc founding classifier, it is deferred to future confirmatory work using an independently assembled founding-site list. The preliminary figures are available in the public repository.

7.5 Conclusion

This study began with a specific metrological hypothesis: that longitudes of UNESCO World Heritage sites encode a 0.1-beru harmonic grid anchored in the Levant, motivated by cuneiform evidence that the Babylonian beru was used as a unit of geographic arc. All results are exploratory; the circularity and analytic consequences are defined in §1.1.

The UNESCO longitude distribution appears to contain a global 3° periodic structure (Test 1, descriptive). All 120 anchors of the form x.18°E yield the maximum observed A+ count, and sub-percent shifts (±0.3%) from the canonical 0.1-beru value collapse significance in both the full corpus and the dome sub-population. The 3° spacing corresponds exactly to 0.1 beru, the smallest integer sub-division of the Babylonian beru unit. The dome sub-population (spherical/domed monuments; **primary pre-specified Test 2**) survives all three simulation nulls, and A+-bearing harmonics host 4.19× more total sites than non-A+-bearing harmonics — a cluster asymmetry that persists under permutation ($p = 0.0033$; **primary pre-specified Test 3**). Multiple-comparisons bounds (Holm and Bonferroni) are applied to the $k = 2$ pre-specified family (Tests 2 and 3 only).

These findings admit a geographic-concentration alternative. A sharpness permutation test of the sensitivity slope (§5.2) shows that the ±0.3% sensitivity peak is *not* unusual under the empirical null ($p_{\text{perm}} = 0.1728$, ns). A stronger specificity test (§5.2) asks whether the peak lands specifically at the metrologically attested 0.1000-beru unit: $p = 0.1728$ (ns). The observed unit-sensitivity pattern is consistent with, but does not independently confirm, the metrological hypothesis. Geographic concentration of heritage sites in Eurasian longitude bands is sufficient to reproduce this feature of the data. The geographic-concentration account must then explain why the longitude clustering of the world's most significant monuments coincides with a metrologically attested unit from ancient Babylonian astronomy. This paper documents the quantitative regularity; its cause is an open question for future investigation.

The two primary pre-specified findings most resistant to a purely geographic account are morphological specificity and cluster asymmetry. For morphological specificity (primary pre-specified Test 2): the primary permutation null confirms dome enrichment ($Z = 3.20$, $p = 0.004$), and the dedicated geographic-concentration simulation (§4.1) establishes that while the dome sites' own longitude positions encode the enrichment (Null A, $p = 0.5514$, ns), other monuments from the same longitude footprint show significantly less enrichment (Null C, $p = 0.0086$, **). For cluster asymmetry (primary pre-specified Test 3): A+-bearing harmonics host 4.19× more total sites ($p = 0.0033$). Geographic clustering does not straightforwardly predict either of these morphological-specificity or harmonic-concentration patterns. Taken together, these two primary observations alongside the global 3° periodicity (of which the unit sensitivity sweep is a second view, not an independent finding) constitute a coherent exploratory pattern consistent with a 0.1-beru harmonic grid awaiting independent historical confirmation.

Criteria for confirmation

Independent replication using a non-UNESCO heritage corpus assembled without reference to this hypothesis would provide the most direct statistical test. The hypothesis would gain substantially from at least one of three forms of independent evidence: (a) cuneiform or other ancient texts explicitly assigning beru-longitude coordinates to distant sites; (b) archaeological survey instruments calibrated in beru sub-divisions; or (c) textual evidence from Hellenistic, Sanskrit, or Islamic geographic sources referencing a Levantine-

centered angular reference frame. Absent such evidence, the pattern documented here is a global statistical regularity consistent with the metrological hypothesis but awaiting independent historical confirmation. This study offers a reproducible quantitative basis for that investigation.

A Keyword Audit Summary

This appendix summarizes the audit trail for the dome/spherical monument filter (Test 2) and the founding-origin classifier (Test E). The full site-by-site audit, including exact UNESCO sentences, gate decisions, and rejection rationale for all dome/spherical sites and all 296 founding-classified sites, is deposited as online supplementary material and is reproducible by running:

```
python3 analysis/unesco/spherical_monument_raw_sweep.py
python3 analysis/unesco/spherical_monument_test.py --audit
```

Note to reviewers: The full site-by-site supplementary audit file is prepared and available upon request. It will be formally deposited to the journal's supplementary data repository at submission. The Zenodo archive (DOI placeholder in the bibliography) will be finalized before submission; the current GitHub repository contains the complete analysis pipeline and all intermediate outputs.

A.1 Dome/Spherical Monument Filter (Test 2 and Test 2x)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text.

Test 2 (primary, population mechanically defined by keyword list): All 90 matching sites are included without any context gating. This raw positive sweep uses a population fully determined by the keyword list alone, with no post-hoc filtering decisions, giving it the strongest methodological standing of the two versions. Lumbini, the Birthplace of the Lord Buddha, is included through the *stupas* keyword, which matches an explicit reference to the commemorative stupa tradition at the site; it is not excluded because the keyword match is architectural and the inclusion rule admits all positive matches without longitude-based decisions.

Test 2x (Exploratory robustness check): Unambiguous keywords (*stupa, stupas, tholos*) are accepted without context validation. Ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns covering geology, vernacular housing, non-architectural objects, and war memorials) rejects non-architectural uses, while Gate 2 (86 positive patterns covering building types, architectural elements, construction terms, and period/style markers) requires architectural confirmation. A site is included if at least one matching sentence passes both gates. Full gate lists: `lib/dome_filter.py`.

Of the 90 raw-sweep sites, 7 are rejected by context validation in Test 2x: Hiroshima Peace Memorial ("dome" refers to the pre-war industrial exhibition hall; UNESCO inscribes the site solely as a peace memorial, not as architectural heritage), Richtersveld (pastoral shelters), Gyeongju (earthen tumuli, captured separately in Test 2b), Viñales Valley (karst formations), Rock Islands Southern Lagoon (natural coral), Uluru-Kata Tjuṭa (geological monoliths), and Diquís Stone Spheres (carved petrospheres). In every case, the exclusion is driven by the semantic domain of the keyword match, not by any property of the site's longitude. Test 2x yields $N = 83$, $p = 0.0005$, $3.31\times$ enrichment, an equal-or-stronger signal than the raw sweep, confirming that the 7 rejected sites are longitude-diluting false positives, not true hits.

A.2 Founding-Origin Classifier (Test E, Post-Hoc Context)

The founding-origin classifier (`lib/founding_filter.py`) assigns sites to five categories: Founding Capital (F), Sacred Origin (S), Founding Monument (M), Founding Axis (X), and Ancient Continuous Landscape (L). Each category uses unambiguous keywords (accepted on sight) and ambiguous keywords (requiring positive and negative context gates). Full keyword lists: `config.json`.

This classifier was iteratively tuned by examining misclassified sites (§3.10). Because refinement occurred after data inspection, the enrichment result has a positive bias of unknown magnitude and is classified as fully post-hoc. It carries no weight in the primary evidential chain (Tests 2 and 3) but is reported here because it speaks to a substantive question: *what kinds of sites, beyond domed and stupa monuments, tend to fall on* $x.18°E$ *nodes?*

Answer. The classifier finds that the A+ set is enriched not only in morphologically domed structures, but in sites associated with the founding of civilizational and religious traditions: early capitals, sacred origin sites, and founding monuments of the major world religions. The pattern is consistent with the beru-alignment hypothesis, which predicts that the most cosmographically significant locations in the ancient world — the sites that later became the seeds of empires and pilgrimage networks — would be most reliably placed on a cosmographic longitude grid. The post-hoc nature of the classifier means this reading cannot be advanced as independent evidence; it is offered as a direction for confirmatory work using an independently assembled founding-site corpus.

Keyword enrichment result. Applied to the full corpus ($N = 1011$), the classifier finds 25 keyword-founding sites among the 56 A+ sites (44.6%; $N_{\mathrm{kw}} = 296$ keyword-founding sites overall, base rate 29.3%). Fisher exact test: OR = 2.04, $p = 0.009$ *.

Known false negatives (e.g., Florence, Timbuktu) affect A+ and non-A+ sites equally, making the enrichment estimate a conservative lower bound even before any post-hoc inflation is considered. The complete version history of the classifier is preserved in the project's `git` commit log and is auditable at the public repository (https://github.com/SethTenenbaum/gerizim-paper-a).

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Data and Code Availability

All data and analysis code are openly available at https://github.com/SethTenenbaum/gerizim-paper-a. A permanent archival snapshot (v1.0.5) with DOI is deposited at Zenodo.

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